ChatGPT

1. Omnibus P-Value Combination Strategy

Combining multiple gene-set test results is central to CATFISH's approach. CATFISH currently uses two omnibus approaches – an **ACAT-O** (Aggregated Cauchy Association Test omnibus) and a **minP-O** (minimum-p omnibus) – to aggregate the five gene-set p-values (Fisher, Stouffer, soft TFisher, ACAT, and minP). This strategy aims to be robust against diverse genetic architectures by allowing any one of the component tests to "win" if it is most appropriate [1]. Below we evaluate this omnibus strategy and potential alternatives:

- **ACAT-O (Cauchy combination)** – ACAT combines p-values via a Cauchy-transformed weighted sum, and has been shown to be *highly robust to dependence among tests while maintaining statistical power* [1]. In rare-variant studies, ACAT-O (omnibus) combining burden, SKAT, and ACAT-V tests outperformed individual tests across genetic architectures [2] [1]. In the CATFISH context, ACAT-O will **prioritize extremely significant p-values** (due to the heavy-tailed Cauchy weighting) but also accumulate evidence from moderately significant results. This makes it effective when *either one test is overwhelmingly significant or when several tests show moderate significance*. ACAT's insensitivity to correlation and ability to combine evidence without permutation is a strength [3].

- **minP-O (minimum p-value)** – Taking the minimum p-value of the five tests (with proper null calibration) is a classic omnibus approach to maximize sensitivity to any single outstanding signal. The minP test is powerful in the extreme scenario where *only one combination method yields a significant signal*. CATFISH's dependence-aware calibration ensures the minP-O maintains correct type-I error under correlated tests (avoiding anticonservativeness that would occur if tests were treated as independent). The downside is that **minP ignores complementary evidence from multiple tests** – it's maximally powerful for a single-dominant-test scenario but can lose power if evidence is distributed across tests.

- **Harmonic Mean p-value (HMP)** – The HMP method is a recently proposed alternative that *combines p-values while controlling FWER* and is particularly powerful in aggregating weak signals [4] [5]. HMP is appealing because it guarantees family-wise error rate control for the omnibus (unlike ACAT which controls only under certain assumptions) [4]. It can detect a group of hypotheses (here, gene-set tests) as significant even if none are individually below the Bonferroni threshold [6]. In practice, HMP tends to outperform standard FDR in power when signals are spread out [5]. **Use case:** if in some pathways each test is only moderately significant (e.g. all five tests yield $p \approx 0.01$), ACAT or minP might not produce a genome-wide significant omnibus, whereas HMP could combine these to a significant result while rigorously controlling false positives [6] [5]. Incorporating HMP as an omnibus (or as an additional combination method) could thus improve power for "broad but weak" signals.

- **Empirical Brown's Method (EBM)** – EBM is an extension of Fisher's method that empirically accounts for dependence among inputs [7]. It has been shown to *outperform Fisher's and other methods under dependence* in simulations [8]. In CATFISH, one might use EBM to combine the five test p-values, or even to replace the classical Fisher test with a correlation-adjusted Fisher. The advantage is **built-in null calibration** using the covariance structure of the p-values [7] [9]. EBM would down-weight the contribution of correlated test results, yielding a more stable omnibus. If the five gene-set methods are moderately correlated (which they likely are, since they derive from the same gene-level data), EBM or related methods (e.g. Kost's method) could provide a *more powerful and correct omnibus p-value than naive Fisher's combination* [8]. This might improve CATFISH's omnibus by capturing dependence explicitly rather than relying solely on simulation.

- **Weighted or Adaptive Combinations** – Instead of treating each gene-set test equally (as ACAT-O and minP-O implicitly do), one could weight them based on prior performance or expected power. For example, if prior simulations show Fisher and Stouffer often correlate with each other, one might *down-weight one to avoid redundancy*. ACAT allows weighting of p-values; CATFISH could assign larger weight to, say, soft TFisher if intermediate sparse signals are deemed more likely, or to Fisher if dense signals are expected. Another idea is **adaptive omnibus selection**: e.g., use the minimum p-value *with a penalty for the number of tests*, akin to Simes' procedure. Simes' method (taking $\min_i {p_{(i)} \times m/i}$) is a valid global null test under positive dependence and could serve as an alternative omnibus combination. It tends to be more powerful than Bonferroni when tests are positively dependent, and it controls FWER in that scenario. However, Simes is essentially a form of adjusted minP and might still be less powerful than ACAT-O in many cases, given ACAT's favorable tail properties.

- **Higher Criticism (HC) and Berk–Jones (BJ)** – These *goodness-of-fit style* combination tests assess whether the distribution of p-values is more concentrated at the low end than expected under null [10] [11]. The Generalized BJ (GBJ) test, in particular, has been adapted for correlation and shown to be powerful for gene-set analysis [11] [12]. BJ effectively finds an optimal threshold τ and tests if the proportion of p-values below τ is larger than null expectation. It is **asymptotically optimal** for certain sparse alternatives – intuitively, BJ (or HC) could outperform fixed methods if, say, ~10–20% of tests show modest significance. In fact, Sun et al. (2019) demonstrated that GBJ offered more power than GHC (generalized HC) and outperformed MAGMA and GSEA in simulations of gene-set association [11] [13]. Incorporating a BJ-type test in CATFISH's repertoire could capture scenarios where *a subset of gene-level signals in the pathway is non-null*, without needing all or just one to be significant. **Recommendation:** Consider adding **GBJ as an additional gene-set test** or even as part of the omnibus. Its analytic p-value calculation for correlated inputs [14] would avoid heavy simulation, and it naturally accounts for pathway size and LD. This could improve power for intermediate polygenicity cases that soft TFisher might not fully capture.

- **Z-score averaging (Stouffer-type)** – Combining p's by averaging Z-scores (with or without weights) is essentially Stouffer's method (which CATFISH already uses). However, one could envision a *cross-test Z combination*: e.g., convert each of the five test p-values into a Z (or t) statistic on a comparable scale and average those. This would treat each gene-set method's result as a "observation" of signal and ask if, on average, the tests indicate significance. Such an approach would be *most powerful if all methods are mildly significant* (because averaging will amplify consistent moderate signals and ignore an outlier). In practice, ACAT-O already achieves a similar goal but with a heavier tail emphasis. A pure Z-average omnibus might be too conservative if one test is highly significant (it would dilute it with the others), so it could miss a strong single-method signal that minP would catch. Therefore, a **weighted Z omnibus** might be preferable only in scenarios where we expect all tests to contribute (for example, if causal signals tend to elevate all combination tests in tandem). Given ACAT's flexibility, a separate Z-averaging omnibus is probably not needed unless empirical testing shows a specific gap in performance.

Cases where alternatives may be preferable:

- If signals are **distributed across many genes in the pathway (dense signals)**, *Fisher, Stouffer, or HMP* are preferable because they effectively sum evidence. ACAT and minP, while still valid, will not downweight moderate signals as aggressively, so their p-values may be less significant in this scenario. For example, if 50% of pathway genes have p≈0.01, Fisher's method will produce an extremely small p-value (summing log-p gives a large test statistic) while minP will still be ~0.01 and ACAT will be on the order of that as well. HMP in this scenario would also yield p ≈ 0.01 (since HMP ≈ the harmonic mean of many 0.01's, which remains 0.01) [6]. Interestingly, HMP's thresholding property means if that 0.01 is below the HMP significance cutoff (which could be less stringent than 0.05 after accounting for tests), it might declare significance where minP would not [5]. So for dense signals, **Fisher or HMP will likely outperform ACAT-O/minP-O**.

- If **only one gene or a very small subset of genes drive the signal (sparse signals)**, then *minP and ACAT* excel. minP is asymptotically optimal for the single-hit case; ACAT likewise gives almost all weight to the smallest p in such cases (e.g. one p=1e-8 and rest ~0.5 yields ACAT p ≈ 1e-8). Fisher's method would also be dominated by that one extremely small p (since $-2\ln(1e-8)$ is huge), so Fisher isn't actually far behind minP in the single-gene extreme. However, Fisher carries extra degrees of freedom (which in a strict analytic test costs some significance) and can be anti-conservative under dependency without adjustment. ACAT's advantage is that it *maintains type-I error even if inputs are dependent*, by virtue of the Cauchy combination's properties [3]. Therefore, for a pathway where one gene test is genome-wide significant and others are null, ACAT-O is a very good choice (and indeed ACAT-O has been shown to be *more robust and powerful than methods focusing only on single-test minima* across various scenarios [1]). In contrast, methods like HMP or Fisher would include the null gene contributions and could require larger sample to reach the same combined significance (though they would still likely be significant in such an extreme case).

- If **a subset of tests show modest signals but others are null**, *soft thresholding or BJ tests* can be preferable. For example, suppose 3 out of 5 gene-set methods produce p ~ 0.001 (indicating some evidence in overlapping manner), but 2 methods produce p ~ 0.5 (no signal). Fisher's method combining all five would be dragged down by the two nulls; Stouffer's average Z similarly diluted. minP would pick 0.001 (ignoring that multiple tests showed it). An adaptive method like BJ might find that, say, "3 out of 5 p-values are ≤0.001" is unlikely under null and yield a significant omnibus. Soft TFisher explicitly is designed for this type of scenario by truncating or down-weighting higher p-values. In fact, *TFisher has been shown to improve power over Fisher for both rare and dense signal cases by tuning the truncation threshold* [15]. If CATFISH's **soft TFisher** is implemented appropriately (with an optimal or adjustable threshold), it likely already covers this scenario well. A BJ test could further improve this by *choosing the optimal cutoff adaptively per pathway*, rather than using one fixed truncation level [10] [12]. Thus, in cases of **intermediate sparsity**, adding BJ might be beneficial, but soft TFisher (if "soft" means it effectively integrates signals below a certain percentile with weights) might capture much of that benefit.

- If **all five tests yield very similar p-values** (which could happen if they are all responding to the same underlying signal in slightly different ways), then any reasonable combination will reflect that value. For instance, if a pathway has *all gene-set tests p ≈ 0.005*, the ACAT-O result will be in the same ballpark (because the Cauchy sum of five identical small p's yields a p of the same order). The minP-O obviously is 0.005 in that case. Fisher's method would give p ≈ 0.000001 (since combining five 0.005

p's yields $\chi^2_{10} \approx -2\sum\ln(0.005)$ which is quite large), while Stouffer would yield p ~0.0001. There is a **discrepancy**: Fisher or Stouffer would signal that collectively these tests are extremely significant (since it's unlikely under null for all to be 0.5% level), whereas minP/ACAT might not register as strongly. This suggests that *when evidence is spread evenly across tests, an additive combination (Fisher/Stouffer) is more sensitive*. ACAT-O might under-weight the scenario of *concordant modest signals*. In such cases, using an **HMP or Fisher-type omnibus** would be preferable. Notably, HMP is constructed to detect joint significance of a set of hypotheses even if none is below α on its own [16]. Thus, CATFISH's reliance on ACAT-O and minP-O could miss pathways where *all tests agree on moderate significance*. One potential improvement is to include an **HMP omnibus**: e.g., combine the five test p-values with HMP and use that as an additional omnibus p-value. HMP would declare the omnibus significant if the joint evidence meets FWER $\alpha=0.05$ (it automatically provides a threshold for significance) [17] [18]. This might catch pathways that ACAT-O/minP-O rank just shy of significance.

Could CATFISH's omnibus be improved? Overall, ACAT-O and minP-O provide a strong one-two punch for extremes of signal aggregation. However, **there is room for refinement**:

- *Consider a hybrid or sequential omnibus:* For example, one might first apply ACAT-O. If the ACAT-O p-value is not significant but multiple tests have moderate p (~0.01–0.05), then apply a Fisher or HMP combination as a secondary test for "distributed signal." This ensures that scenarios where ACAT-O fails (due to no single extremely low p) but cumulative evidence exists are still captured. This two-stage approach could be calibrated to maintain proper Type-I error (e.g., by a Bonferroni of 2 for the two omnibus checks, or by simulation).

- *Leverage ACAT's weighting ability:* ACAT can take weights for each p-value [19]. If prior simulation or theory suggests one test is often superior under certain conditions, CATFISH could assign higher weight to that test's p-value in the ACAT-O combination. For instance, if soft TFisher is thought to cover a middle ground that overlaps with both Fisher and minP, perhaps ACAT-O can drop soft TFisher (to reduce correlation among inputs) or weight it less, focusing on the more "orthogonal" tests. On the flip side, one might weight **minP's p-value less** in ACAT-O because minP is highly correlated with ACAT's own emphasis on the smallest p. Reducing redundancy in the input p-values can make the omnibus more stable. This would require careful tuning to avoid losing power.

- *Add GBJ (Generalized Berk–Jones) to the test battery:* Instead of relying on soft TFisher (whose performance will depend on the chosen truncation parameter), one could include the GBJ statistic for each pathway. GBJ analytically finds the most deviant tail fraction of gene-level p's (accounting for LD) [10] [14]. If added, CATFISH would have six tests: ACAT, Fisher, Stouffer, soft TFisher, minP, and GBJ. An ACAT-O or minP-O could then combine all six. Because GBJ has high power for cases with an "appreciable subset" of genes showing signals [20], it might capture pathways that the current five tests miss. Notably, Sun et al. demonstrated GBJ can outperform GSEA and MAGMA's set test in detecting pathways with distributed signals [13]. The downside is that GBJ's distribution under correlation is complex – but the authors provide analytic p-values [21], so it wouldn't increase simulation burden. Given that CATFISH already employs omnibus calibration by simulation, adding one more correlated test won't complicate matters much; it could simply be folded into the existing null resampling.

- *Drop or replace redundant tests:* The inclusion of both **soft TFisher and ACAT** might be redundant. **Soft TFisher** (a truncated or weighted Fisher) likely overlaps with what ACAT and Fisher together offer. ACAT's Cauchy combination can be seen as a smooth truncation – it heavily weights small p-values but still accumulates contributions from larger p's. Soft TFisher does similarly by truncation or down-weighting. If internal analyses find that soft TFisher's p-values are highly correlated with either Fisher's or ACAT's p-values for most pathways, it might not be adding unique power. For example, if soft TFisher always tends to be in between Fisher and minP, then ACAT (which essentially is an adaptive intermediate) might already cover that range. In that case, **replacing soft TFisher with a more distinct test** (like BJ/GBJ or a non-parametric KS-type test) could broaden the power profile of CATFISH.

- *Directional Stouffer's test:* If there is reason to believe that in some traits, *risk-increasing vs risk-decreasing genetic effects segregate by pathway*, a directional test could be valuable. A **directional Stouffer** would combine gene Z-scores with their sign (e.g., positive Z if the gene's risk allele increases trait, negative if decreases trait). This tests whether the pathway's genes have a consistent direction of effect. For example, in an Alzheimer's disease pathway, one might expect all risk genes in an amyloid pathway to have effects that increase disease risk (positive direction). A directional test would yield a more significant result if indeed most gene effects align (either all $Z > 0$ or all $Z < 0$), whereas a two-sided test might treat opposing effects as mere "significant p-values" and still report enrichment. **When might this be preferable?** – If biological hypotheses suggest a coordinated effect (e.g., a pathway is up-regulated to cause disease, so all causal genes act in the same direction on the phenotype). In practice, however, gene-level association p-values are two-sided and do not carry sign information by default. One would have to assign a direction to each gene (perhaps by the sign of the top SNP effect or a MAGMA gene-score effect estimate). This can be noisy. Moreover, many complex traits have mixed directional effects in pathways (protective vs deleterious variants in different genes) unless there is a clear biochemical cascade. As a result, **directional tests are not commonly used in GWAS pathway analysis** – they are more common in transcriptomics (up/down regulated gene sets). If CATFISH's developers want to explore it, they could implement both a "positive-direction Stouffer" and a "negative-direction Stouffer" for each pathway, then take the minimum of those two p-values as the pathway's directional enrichment. This doubles the number of tests (and the correlation between positive/negative tests would be high because they use largely the same data, just flipping signs). It may be more prudent to hold this in reserve unless a specific scenario (like all risk variants are gain-of-function in a pathway) is suspected. In summary, *directional Stouffer could improve power for pathways with coherent effect direction, but it risks losing power or confusing interpretation when directions are mixed*. A safer approach is to rely on two-sided combination tests (as currently) and handle direction in post-hoc interpretation of gene effects. If evidence of consistent direction emerges (e.g., all top genes in a pathway have the same sign of effect), one can note that rather than incorporate it into the test itself.

In conclusion, the omnibus strategy in CATFISH is quite comprehensive by already using a *dependence-corrected ACAT-O and minP-O*. These cover most extremes – ACAT-O for broad robustness and minP-O for single-strong signals. **To further enhance it**, CATFISH could incorporate an **HMP-based combination** (to capture moderate but consistent signals with strict error control) or the **Generalized Berk–Jones test** (to adaptively target the optimal fraction of significant genes). At minimum, CATFISH's authors should ensure that the chosen five tests are sufficiently diverse in what alternatives they represent. If redundancy is found (e.g., soft TFisher vs ACAT), they might prune one and replace it with a method targeting a scenario not already covered (for example, a **higher-criticism style test for numerous tiny signals** or a **weighted test emphasizing gene-level covariates like gene size**).

Finally, whichever omnibus approach is used, it must be coupled with the **dependence-aware calibration** (discussed next) to maintain validity. None of the combination methods above (except HMP and analytic GBJ) guarantee correct type-I error under arbitrary dependence, so CATFISH's simulation calibration remains crucial [22]. The take-home message is that the omnibus strategy should balance the ability to detect a single gene-driven pathway and a pathway with many weak signals; ACAT-O + minP-O is a good start, but adding **a method for the "in-between" cases** (or using an adaptive approach like aSPU's concept in gene-set context) could further improve sensitivity.

2. MVN Null Calibration and Dependence Handling

A major innovation in CATFISH is its *dependence-aware p-value calibration* using a multivariate normal (MVN) approximation. This accounts for correlations among gene-level test statistics within a pathway, ensuring that gene-set p-values (for each combination method and the omnibus) follow the correct null distribution. We evaluate key aspects of this approach:

Estimating the gene–gene correlation matrix (R_S):
Under the null (no true associations), the vector of gene-level test statistics in a pathway can be modeled as **multivariate normal, $Z \sim MVN(0, R_S)$**, where R_S captures the correlation between Z_i and Z_j for genes i,j [23]. This assumption is widely used in set-based tests; for example, generalized higher-criticism and Berk–Jones tests rely on an MVN of SNP-level or gene-level summary statistics with known covariance (from LD) [23] [24]. The accuracy of CATFISH's null calibration hinges on how well R_S is estimated:

- *Linkage Disequilibrium (LD) Based Approach:* Typically, gene-level association statistics (e.g., MAGMA Z-scores or $-\log p$) are correlated because genes can share SNPs or lie in linked regions. One approach is to derive R_S from SNP correlations. For example, if gene i's test is essentially a linear combination of SNP z-scores in gene i, and similarly for gene j, one can compute $\mathrm{Cov}(Z_i, Z_j)$ by integrating SNP LD between the two genes' SNP sets. MAGMA itself uses a multiple regression model to compute gene scores, which inherently accounts for LD among SNPs within a gene [25]. However, correlation *between genes* (especially in the same pathway or nearby on the genome) must be separately accounted for [26] [27]. A practical way is:
- Use a reference panel (e.g., 1000 Genomes or UK Biobank samples) to calculate SNP correlation (LD). Then for each pair of genes in a pathway, calculate the overlap or linkage between their SNP sets. If genes are on different chromosomes or >~5–10 Mb apart, $R_S(i,j) \approx 0$ can be assumed (negligible LD). If genes are close or overlapping, their test statistics will be correlated.
- Alternatively, one can simulate *null phenotypes* on a genotype panel and derive empirical gene Z-scores to directly estimate correlations. For instance, pick a large number of individuals' genotypes, simulate phenotype with no genotype effect, run MAGMA gene analysis, and compute the Pearson correlation between gene i and j scores over simulations. This empirical R_S would naturally include LD and any gene size effects. This is essentially what Pascal's "gene-fusion" approach avoids by analytically merging correlated genes, but one can do it explicitly [27] [28].

In practice, CATFISH likely uses a more direct route: since MAGMA gene scores are derived from SNP summary statistics, one can compute R_S analytically if MAGMA provides variance–covariance output. If not, computing it via reference LD and gene window overlap is a solid approach. It's crucial that the reference panel's ancestry matches the GWAS data ancestry; otherwise, LD may differ and correlation estimates be off. Studies have shown that using a well-matched reference (e.g., 1000G EUR for European GWAS) yields highly concordant gene results.

- *Handling near-collinearity and overlaps:* If two genes share many of the same SNPs (or one gene's window is fully inside another's), their test statistics can be almost perfectly correlated. This can cause R_S to be singular or nearly so. **CATFISH should detect and handle such cases**. One solution (used by Pascal) is to **merge highly correlated genes** into a single "fusion gene" before testing [26] [27]. Pascal's gene-fusion strategy recomputes a combined gene score for sets of genes in high LD, ensuring the null gene-set p-values remain uniform [28]. CATFISH could adopt a similar pre-processing: e.g., if any pair of genes in a pathway has $r > 0.9$ in R_S, merge them (or drop one if it's essentially a duplicate signal). This prevents double counting of essentially the same locus. Empirical results from Pascal show that without merging, pathways with many neighboring genes show *inflated false positives*, while merging restores the null distribution to uniform [28] [29]. For example, pathway p-values were strongly anti-conservative when genes in LD were treated as separate, but after gene-fusion the inflation disappeared [28]. We recommend CATFISH implement a similar check: e.g., perform hierarchical clustering on genes by LD and fuse clusters (or at least flag pathways with a large fraction of genes in high LD). If merging is too complex, at minimum define each gene's window to avoid large overlaps (e.g., assign SNPs uniquely to the nearest gene or truncate overlapping regions). MAGENTA (an older tool) took the approach of pruning adjacent genes, but that biased towards picking the most significant gene [26] [30]. A full recompute (fusing genes' SNP sets and recalculating a joint gene score) is more principled and retains information [27].

- *Beyond LD – pathway-specific correlations:* Note that R_S might also capture correlation induced by **pathway annotation overlap**. For example, if the same gene appears in two different "functional categories" being tested simultaneously (like gene pleiotropy across pathways), that can correlate test results across pathways. However, within a single pathway analysis, this isn't an issue (each pathway's analysis deals with each gene once). Another subtle source: confounders (like gene size or gene density) – larger genes have more SNPs, which can lead to both higher chance of association and possibly correlation with neighboring large genes. MAGMA's gene model often includes a correction for gene size in competitive testing [25] [31], but in self-contained tests like CATFISH, one must ensure that if gene size strongly influences test statistic variance, it's reflected in R_S or in the simulation. Typically, Z-scores are already standardized so size shouldn't inflate variance systematically, but if any heteroskedasticity remains, an empirical approach could catch it.

Handling missing values in R_S:
"Missing values" could refer to cases where a gene-level p-value is undefined or where correlation cannot be computed. For instance, a gene might have no SNPs after QC or be on the X chromosome when using only autosomal reference, etc. The general best practice is: - If a gene has **no data** (no SNPs or not analyzed by MAGMA), CATFISH should exclude it from the pathway analysis (or treat its p-value as 1, meaning it contributes no signal). In R_S, that gene would simply be omitted (reducing the matrix dimension). - If a gene pair has **no overlapping variants or very distant**, then empirically one might have difficulty estimating correlation (because it's effectively zero within sampling error). In such cases, it's safe to set $R_S(i,j)=0$. - If R_S comes out non-positive-definite due to estimation error or rounding (which can

happen when mixing info sources), CATFISH can apply a slight regularization: e.g., eigen-decompose R_S and set any negative eigenvalues to a tiny positive number. This yields a PSD matrix that is close to the original. - Missing entries (NA) in correlation (perhaps if a gene's data is missing entirely) should be filled with 0 or the gene dropped.

In summary, **each pathway's gene correlation matrix must be clean and invertible** (for MVN simulation or analytic integration). Merging correlated genes helps; dropping genes with no data (or p=1 placeholder) is also important.

Uniform vs empirical marginals for simulation:
Once R_S is set, we need to simulate gene-level test statistics under null. A critical choice is the *marginal distribution* of each gene's statistic: - If we assume under H_0 each gene's test statistic $Z_i \sim N(0,1)$, that implies each gene's two-sided p-value $P_i \sim \mathcal{U}(0,1)$ (uniform on [0,1]). **This is true if the gene's null distribution is well-calibrated**. MAGMA gene p-values, for instance, should be uniform under no association if model assumptions hold. Using the theoretical Uniform(0,1) as marginals is convenient – one can simulate $Z \sim MVN(0,R_S)$, then convert to p-values $p_i = 2(1 - \Phi(|Z_i|))$ for two-sided tests. Alternatively, simulate directly uniform p's with the desired rank correlation structure (via a copula). - However, there might be slight deviations in practice: e.g., discrete SNP count effects, or the test being conservative for genes with few SNPs, etc. An **empirical marginal** approach means we derive the distribution of each gene's test statistic from data or extensive null simulations, and use that instead of N(0,1). For example, one could fit a mixture or use the empirical cumulative distribution function (ECDF) of observed null p for each gene. But since each gene is unique, this is not straightforward (observing a gene's "null" many times requires simulation anyway). - A compromise is to adjust for genomic inflation or other global factors: if the gene-level test statistics have a slight inflation (say $\lambda_{GC}=1.05$), one could scale R_S or Z accordingly or use a heavier-tailed marginal. But ideally, if MAGMA was run with proper covariate control, gene p's should be near-uniform genome-wide under null.

In high-throughput contexts, *Empirical Brown's method (EBM) uses the ECDF of the data matrix to estimate covariance* [32]. CATFISH's approach is similar in spirit: use actual genotype correlations rather than assume independence. We suggest: - **Start with Uniform marginals (Normal Z)** for simplicity, but validate that assumption. One can check if, say, the distribution of pathway p-values under phenotype permutation is uniform. If consistent skew is found, then an empirical tweak is needed. - If using **copula simulation**: generate multivariate normal $(Z_1,\dots,Z_m)$ with cov R_S, then apply the normal CDF Φ to each component to get Uniform(0,1) values $U_i = \Phi(Z_i)$. Those U_i can be interpreted as simulated gene p-values under null, preserving rank correlation R_S. This is effectively simulating from a Gaussian copula with uniform margins – a valid approach as long as the gene test statistic distribution is approximately symmetric and continuous. - If particularly concerned about *gene-specific null distributions* (say a gene with very few SNPs might have a conservative p distribution because test DF is low), one could calibrate that gene's marginal by simulation. For example, simulate null Z for that gene by random phenotype or using MAGMA's internal null model. One could then replace the assumption $Z_i\sim N(0,1)$ with a sampled or empirical distribution for Z_i. But doing this for every gene in every pathway is burdensome and likely unnecessary unless there's evidence of miscalibration.

Recalibrating component p-values vs only omnibus:
This is a crucial question: should each of the five component gene-set tests be individually adjusted for gene correlation, or is it sufficient to adjust the final omnibus p-value?

- *Ideal scenario:* We would obtain **adjusted p-values for each test** (Fisher, Stouffer, etc) that account for the effective correlation among gene p-values in that pathway. For example, Fisher's combination of m gene p-values under independence follows χ^2_{2m}, but with correlated genes the null distribution is stochastically "heavier-tailed" (more small p's than χ^2 expects), leading to anticonservative p's [33] [34]. EBM essentially computes an adjusted Fisher p by using the covariance of $\ln p$ values [32]. Similarly, Stouffer's $Z = \sum w_i Z_i / \sqrt{\sum w_i^2}$ has $\mathcal{N}(0,1)$ under independence, but under correlation its variance is $\sum_{ij} w_i w_j \rho_{ij}$, which if $\neq\sum w_i^2$ will make the naive p off. One could derive an adjusted variance to get a corrected p. The minP raw p should be multiplied by the effective number of tests (like a Sidak correction) if independent, but with dependence one often uses a simulation or a formula from extreme value theory. ACAT, interestingly, *is invariant to certain dependence structures* and often retains type-I error without adjustment (this is a property noted in ACAT literature: as long as the p-values are uniform, ACAT's combined statistic is valid even if they are dependent, thanks to the Cauchy sum's properties) – though extreme correlations can still pose issues. Still, ACAT is generally quite robust [3], so one might skip individual calibration for ACAT combined p.

- **CATFISH's current approach** (implied by "dependence-aware omnibus calibration") likely generates *null samples of all five test statistics together*, then computes the omnibus p (ACAT-O or minP-O) empirically. This means the **final omnibus p-value is correctly calibrated**, but the intermediate test p-values might remain slightly misspecified. For example, CATFISH might report a pathway's Fisher combination p as $1e-4$ nominal, whereas the true adjusted p (taking gene correlations into account) might be $2e-4$. If the omnibus uses all tests together in simulation, it doesn't explicitly correct each one, but it doesn't need to – it just needs the *joint null distribution* of all tests. The question is whether there is value in recalibrating each component:

- Recalibrating each test would allow one to *interpret the individual test results more confidently* and possibly compare across pathways. It might also slightly change the omnibus result: for instance, feeding corrected Fisher p-values into ACAT-O vs uncorrected might differ. However, if we ultimately rely on the omnibus, the relative ordering of test statistics under null vs alternative is more important than their absolute calibration.
- Only calibrating the omnibus (ACAT-O, minP-O) means we trust the final p-value but we know, for example, that the raw Fisher p in a given pathway might be liberal. Since CATFISH's goal is to publish a method, they may present the five test p-values for insight, so it could be worth calibrating them too.

Best practice: *Calibrate at least the omnibus, and if computationally feasible, also calibrate the individual test p-values.* This can be done by utilizing the same MVN simulation: - Perform B simulations of the gene-level Z vector for the pathway. - For each sim, compute all five combination test statistics (Fisher's X^2, Stouffer's Z, etc.) and also the omnibus stats (ACAT-O Cauchy sum, minP). - The empirical distribution of each test statistic gives its null CDF. From that, derive an adjusted p-value for the observed statistic. For instance, if out of B simulations, the Fisher combination statistic was as extreme as observed in k cases, the adjusted p $\approx (k+1)/(B+1)$. - Do similarly for omnibus: e.g., for ACAT-O, take the combined Cauchy statistic in each sim, etc.

By doing this, **one simulation serves to calibrate everything**. There is practically little extra cost to calibrate the components once you are simulating the joint null, except bookkeeping. This yields dependence-adjusted p-values for Fisher, Stouffer, etc., which could be useful to report. For example, one might find that *the "minP" test unadjusted p was $5e-6$ but after accounting for gene–gene LD it's $1e-4$*, highlighting how LD inflation was making minP overly optimistic.

However, one might also choose a middle ground: *only calibrate the omnibus* because that's what determines significance, and treat component test results qualitatively. If the simulation budget is limited (say $B=100,000$), tail estimation for each test could be noisy. Focusing on the omnibus alone means you can put more effort (maybe $B=1e6$) into that single p-value. Given that CATFISH's "omnibus p-value calibration strategy" is emphasized, I suspect the primary output is the omnibus p for each pathway, which they ensure is accurate via simulation.

Null simulation strategies (pathway-specific vs genome-wide):

- **Pathway-specific MVN simulation:** The straightforward approach is to simulate null gene stats for one pathway at a time using its R_S. E.g., for a pathway with 20 genes, generate B samples from $MVN(0,R_{20x20})$, compute the five test p's and omnibus for each, and thereby estimate p-values. This must be repeated for each pathway. If there are thousands of pathways, this is computationally heavy but embarrassingly parallel (each pathway's sim is independent). The accuracy in the tail (e.g., $p<10^{-6}$) depends on B. One can choose B adaptively per pathway: if a pathway's observed p is modest (say 10^{-3}), a smaller B (like $1e5$) suffices for precise calibration. If observed p is extreme (10^{-6}), one might need $B=10^7$ for stable Monte Carlo estimate. Running very large B for every pathway is impractical, so **adaptive resampling** is recommended:
- Run an initial simulation with B_1 (e.g., 10^5) for all pathways to get rough p's.
- Identify pathways that appear significant or close to threshold.
- For those, run additional simulations (total B_2, e.g. 10^6 or more) to firm up the tail probability.
- Possibly combine variance by importance sampling (e.g., oversample tail events).

The computational burden could be mitigated by the fact that many pathways will not be close to significant, so they don't need super-precise tail estimates. They will have, say, p ~0.5 or 0.1, which can be estimated with less effort. Focus simulation where it matters.

- **Genome-wide (or multi-pathway) resampling:** An alternative is to simulate *the entire genome's gene Z-scores* in one go, then derive all pathways' test stats in each simulation. This has an appeal: you generate one null GWAS outcome and run the full pipeline (compute MAGMA gene p's and then pathway p's) – essentially a *permutation test* approach at the pathway level. Doing this N times yields an empirical null distribution for *the number of significant pathways* or the minimum p observed, etc. Westfall–Young step-down correction often uses such joint resampling to account for correlation between tests. The advantage is you can directly estimate the *family-wise error*: e.g., count in how many of the N null replicates any pathway had p < some threshold. This gives an empirical FWER and allows setting a cutoff without assuming independence of pathways. The downside is simulating a full genome MVN of gene statistics is extremely high-dimensional (e.g., 20,000 genes). Even with tricks (block by chromosome), it's heavy. Also, deriving pathway p's from a full null GWAS requires recomputation of combination tests which is doable but not trivial for each replicate.

Practically, CATFISH likely sticks to *pathway-specific calibration for each pathway's p-value*, then separately addresses multiple testing across pathways via FDR/FWER procedures (see section 3). This is sensible because most pathway tests are approximately independent or only modestly correlated except via overlapping genes. Those overlaps could be handled by a second-layer adjustment if needed (like effective number of pathways).

- **Resampling strategies:** If individual gene correlations are well-captured by R_S, simulating from MVN is straightforward and fast (drawing from 20x20 or 50x50 covariance is trivial). The challenge is the tail: with, say, $B=10^5$, the smallest resolvable p-value is ~1e-5 (one observation in tail). If one needs $1e-6$ accuracy, B must be larger or combined with extrapolation. Here are some strategies:
- *Importance Sampling:* e.g., to estimate extremely small p for minP, one could sample conditional on one gene being in an extreme tail. But for an arbitrary combination test, designing a biasing distribution is complex. Another approach is to use a theoretical extreme value distribution (if available) to fit the tail. For instance, for minP one could use the Sidak correction or generalized extreme value (GEV) distribution to model the minimum's tail with parameters estimated from intermediate quantiles.
- *Empirical fitting:* Run a reasonably large simulation (e.g. 10^6), and if no event surpasses the observed statistic, use the largest few simulated values to fit, say, a generalized Pareto distribution (GPD) for the tail beyond a threshold. Then extrapolate to the observed level. This can give an estimate of p beyond simulation resolution.

- Given the complexity, a conservative approach is to ensure B is large enough that a handful of sim values exceed the observed statistic for significant pathways. For example, if aiming for genome-wide $\alpha = 5e-6$, one might target $B=1e7$ for those top pathways, expecting about 50 under null at p=5e-6.

- **Choice of B for accurate tail:** The required B grows inversely with the p-value threshold of interest. If one wants accuracy at $\sim 1e-6$ (the order of genome-wide significance for ~10k tests), B on the order of 10^7–10^8 might be needed to get a stable estimate [22]. Sun et al. (2019) pointed out that *permutation* to control FWER at 0.05 across 10k pathways would require ~10 million iterations [22]. Monte Carlo simulation is similar. That is a lot, but perhaps manageable for a single pathway on a cluster – not for thousands. Thus, again, focusing on only the top signals with high B and using analytic approximations for others is pragmatic:

- Many pathways will have p in the range 0.001–1. For those, $B=100k$ (with a minimum resolvable p of ~1e-5) is fine, since we don't need more precision – they're not close to $5e-6$ anyway.
- For the handful of pathways that *appear* to be $p<1e-4$, progressively increase B until the p-value estimate stabilizes or crosses the significance threshold.
- If computation allows, do a *bulk simulation* of moderate size (e.g., $B=1e5$) for all, then a *targeted simulation* for top 5% of pathways with $B=1e6$, etc.

To summarize, **pathway-specific MVN simulation with an adaptive B scheme** is recommended. Ensure R_S is accurate (which may involve merging genes or high-quality LD data), use Uniform(0,1) marginals via a Gaussian copula (which assumes gene test statistics are z-scores under null), and validate that by a few phenotype permutations if possible. This will yield calibrated p-values for each pathway's tests and omnibus. The MVN approach is efficient and has been shown in other tools (like Pascal and GBJ) to preserve type I error at strict levels [28] [21].

One more consideration: **should one simulate under a pathway-specific null or global null?** There's a nuance – if the *gene set tests* have covariates (like MAGMA competitive analysis adjusts for gene size, gene density, etc.), those covariates might induce slight differences in null distribution per pathway. CATFISH's gene-set tests are self-contained (just the genes in the set), so the null is just "no gene has effect." Simulating under that condition for each pathway separately is correct. If one were to simulate the whole genome, one implicitly assumes no pathway has effect (global null), which is fine for error control. Pathway-specific simulation is essentially the same assumption but done individually. No major difference except efficiency.

Component p-value recalibration vs omnibus only – which is better?
We lean towards *recalibrating only the omnibus* if resources are constrained, because ultimately our decision of significance is based on the omnibus p. However, presenting well-calibrated component p-values can provide insight into *how* a pathway is significant (e.g., "Fisher's test (corrected) p = 1e-3, minP test p = 5e-5, omnibus p = 2e-5"), which readers might find useful. Given that a single unified simulation can give all of these, CATFISH should consider doing so. If simulation error for component tests is a worry, note that those component tests typically have less extreme p's than the omnibus (the omnibus is designed to be smallest or among the smallest), so they are easier to estimate.

In practice, some methods choose to calibrate the omnibus by simulation and rely on asymptotic approximations for each component test (to save time). For example, one might use Brown's adjusted Fisher p analytically [32], use a simple moment-based correction for Stouffer (i.e., use $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \mathbf{w}^T R_S \mathbf{w}$), and use known bounds for minP (like Šidák or Dunn). Those adjusted p's could be plugged into ACAT-O, possibly reducing the need for heavy simulation. However, those analytic adjustments are themselves approximations (Brown's method is quite accurate though [34]). CATFISH's fully empirical strategy (via MVN draws) is more general.

Null simulation best practices: - Use high-precision linear algebra for generating correlated normals (e.g., Cholesky or eigen-decomposition of R_S). Verify R_S is PSD; if not, correct it. - Ensure random seeds and parallel streams are handled to get reproducible p-values if needed. - Use a large number of simulations for pathways that matter, but remember no simulation can give p=0, so if an observed p=0 in simulation, report it as $<1/B$ or use an extreme value fit. - It's wise to perform a **sanity check**: simulate an entire null GWAS (all genes MVN), run the whole CATFISH pipeline (maybe for a subset of pathways to save time) and confirm that the distribution of omnibus p-values is uniform (or that, say, the proportion of pathways with $p<0.05$ is ~5%). Pascal did a similar check with random phenotypes and showed uniform QQ-plots with their LD corrections [28] [29]. CATFISH should similarly demonstrate that its calibrated p's are indeed uniform. This builds confidence in the MVN assumptions.

In conclusion, **CATFISH's MVN null calibration approach is appropriate and appears necessary**. Estimating R_S accurately (via LD and gene merging) and using that to simulate null statistics is the gold standard for type I error control in gene-set tests [28] [29]. The marginals can be treated as Uniform(0,1) (equivalently Normal(0,1) for Z-scores) provided MAGMA gene p-values are well-calibrated; any inflation should be corrected at the gene level beforehand (e.g., via GC correction on gene scores if needed). We advise recalibrating at least the *omnibus p-value* per pathway using these simulations. Recalibrating the individual test p-values too is ideal and can be done with minimal overhead during the same simulations. By doing so, CATFISH will ensure not only that the final combined p-value is correct, but also that each constituent test's evidence is properly quantified under correlation. This thorough null modeling will position CATFISH as a reliably calibrated method, in contrast to some earlier methods that relied on permutation and still suffered type I error issues for correlated genes [35] [36].

3. Genome-Wide Significance Thresholds for Pathways

Having obtained calibrated p-values for each pathway's omnibus test, the next challenge is determining which pathways to declare significant in a **genome-wide context**. With potentially thousands of pathways tested, we must control for multiple comparisons. Here we discuss strategies for defining significance thresholds across pathways – comparing **Family-Wise Error Rate (FWER)** control to **False Discovery Rate (FDR)** control – and the use of empirical null simulations to set cutoffs.

FWER Control (e.g., Bonferroni or simulation-based):
FWER at $\alpha = 0.05$ means the probability of even one false positive pathway is at most 5%. This is the most stringent criterion, analogous to the 5×10^{-8} SNP-level threshold in GWAS which controls FWER across ~1 million tests. For pathway analysis, if we have e.g. $M = 10,000$ pathways, a Bonferroni correction would set $\alpha_{\text{pathway}} = 0.05 / 10,000 = 5\times 10^{-6}$ as the significance cutoff for each pathway's p-value. Indeed, many pathway studies use such thresholds when seeking very high confidence results. For example, Lamparter et al. (2016) in Pascal considered pathways significant if $p < 5\times10^{-6}$ (0.05/10,000) [37], and described it as analogous to a "genome-wide" threshold.

- *Advantages:* FWER control yields strong claims – any declared pathway has ≤5% chance of being a false alarm under the global null. It avoids the complications of interpreting an expected proportion like FDR. Also, when results are sparse (a few pathways with tiny p's), Bonferroni is not overly punitive.

- *Disadvantages:* It can be **conservative**, especially if tests are not independent. Many pathways in databases are correlated (due to gene overlap); thus the "effective" number of independent tests may be less than the total count. Bonferroni doesn't account for that, potentially underestimating α_{pathway}. For instance, 100 pathways all containing FGFR2 are not 100 independent tests, yet Bonferroni treats them as such (driving the threshold lower than needed) [38] [39]. If one wanted to adjust for that, a Sidak or effective number of tests approach could be tried, but those require calculating the correlation among pathway test results.

- *FWER via simulation (empirical null maxima):* Instead of assuming independence for Bonferroni, one can **simulate the null across all pathways** and measure the distribution of the minimum p-value or the number of pathways below certain p thresholds. As discussed, generating full null replicates of all pathways while accounting for their overlaps is complex but feasible in principle. One could do:

- Simulate gene-level summary stats for the whole genome (or for each chromosome independently),
- Compute all pathway omnibus p-values for each simulation,
- Then find, e.g., the 95th percentile of the minimum p-value across pathways (this is the threshold where in 95% of null simulations, no pathway has a p below that).
- Use that as the genome-wide cutoff.

This approach was conceptually used by some methods; for example, some early gene-set methods performed **permutation of case-control labels** and in each permutation took the minimum p across all pathways to build the empirical distribution of the min (thus controlling FWER). However, as noted, to reach an extreme tail like the min p, one needs many permutations. Sun et al. mentioned that to Bonferroni-correct 10k pathways via permutation, ~10 million permutations would be needed [22], which is computationally prohibitive. With MVN simulation one might do fewer (because you can simulate faster than permuting full GWAS possibly), but still millions are needed for stable extreme quantiles.

An alternative is to estimate the *effective number of independent pathway tests, M_{eff}*. Methods like Li & Ji (2005) for SNPs use eigenvalues of the correlation matrix to get M_{eff}. One could compute a correlation matrix of pathway results (maybe by overlapping genes: pathways sharing many genes will have correlated p-values under null). Then sum of eigenvalues or some formula gives an effective number of tests M_{eff}. Then use α / M_{eff} as threshold. For example, if 10k pathways are largely overlapping, M_{eff} might be, say, 5000 independent tests, making the threshold $1e-5$ instead of $5e-6$. This is a heuristic but commonly used in multiple testing of correlated markers.

Another simulation approach: simulate many null GWAS, record the minimum pathway p each time, and fit an extreme value distribution to those minima. If the minimum p approximately follows a distribution (maybe exponential in log-scale or Gumbel), one could extrapolate to 5% tail with fewer simulations. This is advanced and perhaps overkill given simpler methods like FDR are available.

FDR Control (Benjamini–Hochberg or others):
FDR at 5% means *on average, 5% of the significant pathways are expected to be false positives*. This is a less stringent criterion that allows more discoveries, accepting a controlled proportion of error. FDR is often favored in exploratory omics analyses (like gene expression, GO enrichment) because we expect multiple findings and are willing to tolerate a few false ones as long as the majority are true.

- The standard **Benjamini–Hochberg (BH) procedure** can be applied to the list of omnibus p-values for all pathways. One would sort the pathways by p, find the largest k such that $p_{(k)} \le \frac{k}{M} q$ (with $q=0.05$ for 5%), and declare those k pathways significant at FDR 5%. BH has a theoretical guarantee of controlling FDR under independent or positively correlated tests, and tends to be conservative if there are arbitrary dependencies. Given pathway overlaps can induce positive correlation (pathways sharing genes will tend to both have low p if that gene is truly associated, or both high p if not), BH should be relatively fine (positive regression dependency condition likely holds). If some pathways are very overlapping, BH can be slightly anti-conservative in worst cases, but typically not severely [38].

- FDR is more **lenient**: for example, if there are many pathways with p ~1e-4, BH might call some significant whereas Bonferroni would not until 5e-6. This can be desirable if one wants to cast a wider net for potential pathways to investigate, especially if the cost of follow-up is low (e.g., functional interpretation or further genotyping is not as costly as, say, a new experiment). Many gene-set analysis papers present FDR < 0.05 lists as the main output.

- The interpretation of FDR is slightly different: one expects 5% of reported pathways to be false. If we report 20 pathways at FDR 5%, maybe one could be a false alarm.

Recommendation – FWER vs FDR:
This depends on the goals. If CATFISH is aiming to demonstrate strict *genome-wide significance akin to SNP GWAS*, using FWER (Bonferroni or similar) can make their results comparable in stringency to SNP-level findings. E.g., saying "Pathway X is significant at 5×10^{-6}, analogous to genome-wide significance" has a certain weight. On the other hand, if the emphasis is on discovery and generating hypotheses, FDR is widely accepted and will yield more findings.

It might be wise to use both: Define a **genome-wide significance threshold (FWER-controlled)** for pathways for high-confidence hits, *and* also discuss pathways passing FDR 5% as suggestive or worthy of follow-up. This mirrors what is done in gene expression analysis (e.g., "no pathway survived Bonferroni, but many passed FDR 10%").

In a publication, one could present a Manhattan-plot-like figure of pathway –log10 p-values with a Bonferroni cutoff line (analogous to GWAS). If a pathway exceeds that, it's a "genome-wide significant pathway." Additionally, an enrichment analysis of how many pathways pass FDR could reveal broader trends even if none pass Bonferroni.

Use of empirical null maxima to set cutoffs:
If CATFISH authors want to be very rigorous, they could use the simulation infrastructure to refine the threshold beyond Bonferroni: - For example, simulate 1000 null replicates of all pathway tests (maybe by sampling null gene z's as in section 2 on a genome-wide scale but coarse B). For each replicate, record the minimum omnibus p across all pathways. Sort these 1000 minima. The 50th smallest (i.e., 5th percentile) would be an estimate of the 0.05 FWER threshold. If that turns out to be, say, 8×10^{-6} (just guessing), that suggests Bonferroni was a bit conservative (5e-6 vs 8e-6). With more replicates or some extrapolation, they might zero in on ~7e-6 at 5% FWER, for instance. - Or similarly, estimate how many false positives to expect at a given threshold. For instance, if one picks FDR, one might verify via simulation that actual FWER is under control or measure the false discovery proportion.

Another approach is **Permutation-based FWER control** (Westfall–Young maxT). That's similar to above: permute phenotypes (hence gene associations) multiple times, compute pathway p's, and for each permutation note the smallest p. This gives an empirical distribution of the smallest p under null and one can see where the real data's min p falls. If, say, the real min p is smaller than all permuted mins, you have an estimate p~<1/#perms. One can also derive adjusted p-values for each pathway: the adjusted p is $\Pr(\min_{pathway} p_{\text{null}} \le p_{\text{obs}})$. This method can be more powerful than Bonferroni when tests are correlated, because the permutation null captures the correlation structure. Westfall–Young step-down can further improve power by sequentially adjusting. However, doing this by phenotype permutation is extremely slow for GWAS (which is why methods like MAGMA avoid it [40]). With CATFISH's simulation approach, we can mimic it somewhat faster, but we need many replicates for extreme p's.

Given these considerations: - **If one or few pathways have** omnibus p << 5e-6**, they are robust regardless of method (will pass both FWER and FDR). - If many pathways have modest p-values (say dozens at 10^{-4} to 10^{-6}), then Bonferroni might flag few or none, whereas BH-FDR might flag many. This scenario likely indicates widespread polygenic signal (like many related pathways enriched). In such cases, relying solely on Bonferroni might under-represent the biology. For example, an analysis of height might find *no single pathway passes 5e-6, but an excess of pathways at 1e-4 indicating true enrichment overall*. BH-FDR would reveal that (since if many p ~1e-4, BH will rank them and perhaps call some significant if the enrichment is above null expectation). - There is also the** "enrichment of signal" vs "single outlier" distinction**: Bonferroni cares only about the single most significant pathway (for FWER). FDR cares about the distribution of all p-values. If the entire set of pathways shows more low p's than expected (which might happen if phenotype has pathway-level polygenic basis), FDR will identify a set of significant pathways even if none reached the Bonferroni threshold individually. This is analogous to how in expression studies, many genes can be significant by FDR even if none meet tiny Bonferroni p's.

Setting cutoffs using null simulations of maxima:
As an example, suppose we empirically find that under null, the minimum p across 10k pathways follows approximately a distribution where the 5% quantile is 7×10^{-6} (meaning in 5% of null worlds, at least one pathway had p < $7e-6$). Then using $7e-6$ as threshold would be FWER$\approx$5%. If we used $5e-6$ (Bonferroni), FWER might be 3% in reality (slightly conservative). If we used $1e-5$, FWER might be ~10% (too high). So an empirical approach could fine-tune the threshold. Similarly, one could simulate to find that on average, e.g., there are 0.3 false pathways at $1e-5$ threshold (so ~3% FWER), etc. This level of nuance might not be necessary unless the results are borderline. Usually, Bonferroni is acceptable as a safe threshold. If we wanted to be less stringent, we could justify something like *Bonferroni with effective tests* as above.

Recommendation for publication:
Clearly state what threshold is used and why. For example: - "We considered pathways with omnibus $p<5\times10^{-6}$ as genome-wide significant (FWER 5% level, analogous to SNP GWAS) [22]. This threshold corresponds to a Bonferroni correction for ~10,000 tests and is slightly more conservative than an empirical null simulation which gave ~7×10^{-6} for FWER 5%." – citing if possible the simulation or references. - Additionally, "Pathways with FDR q<0.05 were considered suggestive/significant in a broader sense and are reported for completeness." Using BH on the calibrated p's, list those.

Simulating empirical null maxima to set cutoffs:
If CATFISH already has the machinery to simulate null gene stats, doing a secondary simulation across all pathways could strengthen confidence in their threshold. They might do something like: - Simulate 1000 null outcomes for all pathways (which could be done by doing the gene-level simulation and computing all pathway combos). - Determine the smallest p in each simulation (or the number of pathways below certain p). - If, for instance, no simulated null ever had a pathway p < 10^{-6}, then any real pathway with $p=10^{-8}$ has an empirical $p<0.001$ by that metric – extremely significant. - Conversely, if simulated nulls sometimes produce one or two pathways with $p ~10^{-5}$, then a real pathway with $p=8\times10^{-6}$ might not be that extraordinary in a global sense (maybe FWER ~0.1 in that hypothetical).

However, because of heavy computational cost, a full simulation for FWER might not have been done by the authors. Many rely on Bonferroni as a known conservative bound, which is acceptable and simple.

FWER vs FDR in practice:
In many published pathway analyses: - If one or two pathways pass Bonferroni, they highlight those as "genome-wide significant pathways". - They may also mention if a larger set passes FDR 5%, indicating additional candidates. - If none pass Bonferroni (common), they often fall back to FDR to report something. - If a method's novelty is calibration, they might emphasize using FWER to be rigorous. But showing FDR results too can demonstrate power.

One note: Because CATFISH uses an omnibus that potentially combines correlated tests, the distribution of p-values under the null could deviate from uniform if not calibrated correctly. But we assume they've calibrated well (section 2). So the multiple testing correction can treat the p's as uniform(0,1) i.i.d. for simplicity in BH or Bonferroni calculations, knowing there's some dependency due to overlapping genes. Over-conservatism from that dependency can be counteracted by empirical or effective counting.

Effect of pathway database properties:
The number of pathways and their overlaps matter. For example, if testing **all GO terms (~6000)** plus KEGG, Reactome, etc, you might have M ~ 10-20k. If focusing only on curated pathways (say 300 pathways), thresholds differ (0.05/300 ~1.7e-4). So CATFISH should clarify what database and how many tests. Using an appropriate threshold relative to M is key. Possibly an *FDR might be more stable across different sized pathway sets*, whereas Bonferroni threshold changes drastically if M changes (though one can always quote the number of tests used).

Setting cutoffs via empirical null maxima:
In summary, we support: - Use **Bonferroni (FWER)** for a stringent genome-wide significance bar (e.g., highlight any pathway with $p < 5×10^{-6}$ or whatever is appropriate for M). This gives clear, easy-to-understand criteria (similar to GWAS). - Use **FDR (BH q<0.05)** for broader discovery. Indeed, FDR is arguably more appropriate when many pathways may be truly enriched; FWER can be too strict and miss true positives if they are moderate. BH is simple to apply: sort p's and find the cutoff. - Possibly mention **FWER via simulation** as a validation: e.g., "We confirmed via null simulation that this Bonferroni threshold corresponds to ~5% FWER given the dependency structure of our pathways." If they did that, they can cite those results or else a reference that *permutation-based methods have been recommended to correct pathway p's* [22] [38] but we used Bonferroni for simplicity.

One can also set FDR threshold to achieve something akin to "genome-wide discovery". Daniel Wilson's HMP method actually directly provides FWER control: if one uses HMP grouping all tests by pathway, any pathway with HMP below 0.05 is an FWER-controlled discovery [4]. That's a neat alternative: one could treat each pathway's genes as a "group" and apply the HMP multi-group procedure (which can test each group while controlling overall FWER). Wilson (2019) shows that HMP can test each pathway for association and control FWER across all pathways, often with more power than Bonferroni [5]. It does so by combining p-values of genes in each pathway in a harmonic mean and adjusting via weighting by group size. If CATFISH were to incorporate that approach, it might avoid the need for heavy multiple test correction after the fact. However, implementing that is not trivial and goes beyond the current method (which uses fixed combination tests + simulation calibration). It's an interesting idea but probably out of scope for now. Still, this highlights that *some methods (like HMP) inherently address the across-pathway multiple testing by hierarchically testing groups and subgroups* [41]. In absence of that, we stick with BH and Bonferroni.

Empirical null maxima – additional nuance:
One situation to be cautious of: if *one gene drives many pathways*, then under the true null (no gene associated) this scenario rarely produces many low p's (since no gene is low). But in real data with one gene truly associated, you can get many pathways with low p (all containing that gene). This is a dependency that null simulations (which assume no gene associated) won't capture because it's an alternative scenario. For multiple testing correction, that's fine – we consider controlling error under null. But it means if, say, FGFR2 is associated, it might light up 100 pathways; BH might then call dozens significant with very low q. Bonferroni might only allow one or two. This is not a false positive issue (they're true signals in a sense, but redundant). It's more an interpretation issue (addressed in section 6). For thresholding, one might not want

to “credit” all 100 as independent discoveries. That’s more of an interpretation/presentation problem than a statistical error rate one, though.

Summary: We recommend *reporting genome-wide significance using FWER (Bonferroni or similar) thresholds*, as it gives a direct analogue to SNP-significance and is easy for readers to grasp. For example: “Using a Bonferroni correction for 9,520 pathways tested, we set a significance threshold of 5.25×10^{-6}. Pathways surpassing this are considered significant with FWER<0.05.” Then also perhaps: “Additionally, we report pathways passing FDR q<0.05 as suggestive associations to highlight broader enrichments.” If computationally backed, mention that an empirical null analysis was done: e.g., “Permutation of phenotype labels confirmed that this threshold corresponds to an empirical family-wise error of ~5%” or if not, cite others who have discussed permutation counts [22].

Finally, controlling FWER vs FDR is a choice. If CATFISH is being prepared for publication as a method, they should demonstrate its ability to control type I (likely via FWER demonstration) but also showcase power (likely via FDR or simulation where many are true). They could present both depending on context. As long as the calibrated p-values are accurate, applying BH or Bonferroni is straightforward.

4. Critique of the Five Chosen Gene-Set Tests

CATFISH combines five distinct gene-set test statistics – **Fisher’s method, Stouffer’s method, soft threshold Fisher (soft TFisher), ACAT, and minimum-p (minP)** – before applying the omnibus. Each test captures a different “signal pattern” in the data, which is a strength of the method. We will assess these choices and consider whether adding or substituting other tests (GSEA-KS, Berk–Jones, Higher Criticism, etc.) could improve power, as well as examine overlap/redundancy among the current tests. We also address the idea of using a directional Stouffer’s test.

Recap of current tests and their niches:

- **Fisher’s combination (a.k.a. sum of logs)**: This test sums $-2\sum_{i=1}^m \ln p_i$ for the m genes in the pathway, yielding a χ^2_{2m} under independence. Fisher’s method is *most powerful in “dense” signal scenarios* – i.e., many genes in the set have modest association. It essentially tests if the pathway’s gene p-values as a whole are shifted toward low values. However, Fisher’s can be dominated by a few extremely small p’s as well (since $\ln(p)$ grows large as $p \to 0$). Fisher’s assumes independence, which is violated by gene LD; hence its null distribution was likely corrected by simulation in CATFISH. Fisher’s method (especially after correlation adjustment) is a classic, general-purpose test [7] [34] and is a sensible inclusion for broad sensitivity. Its weakness is in very sparse cases – if only one gene is truly associated, Fisher’s power is less than minP. But if, say, 30% of genes have p ~0.01, Fisher’s will yield a extremely small p-value (leveraging the cumulative evidence).

- **Stouffer’s Z-score method (sum of Z)**: This test typically computes $Z_{\text{set}} = \frac{1}{\sqrt{m}} \sum_{i=1}^m \Phi^{-1}(1-p_i)$ (for equal weights), or a weighted version if gene sizes differ. It is similar to Fisher in that it aggregates all evidence, but by summing Z rather than log p. Intuitively, Stouffer gives more *linear weight* to each gene’s effect. One key difference: an extremely small p (say $p=10^{-8}$) corresponds to $Z\approx 5.5$, while $\ln(p) \approx -18$. So Fisher massively amplifies that single small p, whereas Stouffer adds 5.5 to the sum – noticeable but not overwhelmingly dominating if there are many genes. Thus, **Stouffer's is a bit less sensitive to a single outlier and more sensitive to moderate consistent effects**. In practice, with many genes, Stouffer's statistic will be approximately normal under null if properly accounting for correlations. Stouffer's method is mathematically equivalent to performing a one-sample z-test on the gene Z-scores (testing if their mean is >0). It is most powerful if every gene has a tiny shift in Z (e.g., all genes have a non-null effect of similar magnitude), thus all p's are modestly low. It's also symmetric in that up and down (if using two-sided p's, it takes absolute Z typically). CATFISH likely uses two-sided p's in all combinations (losing sign), so Stouffer is probably combining |Z| or combining one-sided p's for positives and negatives separately might not be done unless they explicitly did directional.

Overlap with Fisher: Both are aggregators. They will often correlate – if many genes have p=0.01, Fisher and Stouffer will both yield significant results. The difference is how they weight tail vs bulk. Fisher (because of log) is more "tail-sensitive" – a few very low p's heavily influence it. Stouffer is more "averaging" – a consistent moderate effect across many genes will register strongly, and it doesn't allow one gene to completely dominate if there are others. In essence, *Stouffer's method is like a less-extreme version of Fisher's*. They are not independent tests by any means; their correlation could be high in many scenarios. However, there are cases where they diverge: - If one gene is extremely significant and others null, Fisher will be more significant than Stouffer (Fisher $\sim$ huge, Stouffer $\sim$ moderate). - If half the genes are at p=0.05 and half at 0.50, Fisher will pick up some signal (log 0.05 = -3, log 0.5=-0.693, sum ~ -3*m/2 - 0.69*m/2, so not huge), whereas Stouffer sums z ~ (1.645 * (m/2) + 0 * (m/2)), giving some moderate Z. Actually both would find something if m is large enough. - If all genes have identical p, then obviously they carry the same info (with mild differences in scaling).

Including both Fisher and Stouffer in CATFISH may introduce redundancy. Many pipeline implementations choose one or the other as their "aggregation test." The benefit of having both is just to be sure – if an alternative scenario arises, maybe one edges out the other. But likely they will largely agree on ranking pathways. We might suspect their inclusion is partly for completeness.

- **soft TFisher (soft threshold Fisher)**: TFisher stands for *Truncated Fisher's method*. The idea of TFisher (Zhang et al. 2020) is to only combine p-values that are below a certain threshold τ, and/or to weight them, to optimize power for intermediate signal proportions [15]. A "hard" truncated Fisher would be: $T = \sum_{i=1}^m -2\ln(p_i)\mathbf{1}(p_i \le \tau)$. This means we ignore all p's above τ (consider them as no information) and only focus on the subset of genes that are fairly significant. This is more powerful than Fisher if a large subset of genes are null (p ~ uniform) and only some fraction are non-null – because including the nulls in a full Fisher just adds noise/degrees of freedom. TFisher can significantly improve power over Fisher in both "rare signals" and "dense signals" regimes by tuning τ [15]. The "soft" TFisher presumably means instead of an indicator cutoff, they use a *weighting function that gradually down-weights larger p-values* instead of outright ignoring them. For example, a soft threshold could be a continuous weight like $w(p) = \min(1, \tau/p)$ or something that approximates giving full weight to $p < \tau$ and fractional to just above τ. The exact implementation might come from the TFisher paper, which allows a combination of truncation and weighting schemes.

Power profile: Soft TFisher (and truncated methods) excel when *only a subset of genes in the pathway are potentially associated*, and one does not want the bulk of null genes to dilute the signal. It is in between minP and Fisher: - If the optimal fraction of causal genes is, say, 10-20% of the pathway, one can choose τ such that roughly that fraction of null p's fall below τ by chance. TFisher will then produce a statistic sensitive to deviation in that fraction. - If the signals are extremely sparse (1 gene), TFisher with a very low $τ$ would approach minP's behavior (it essentially becomes Fisher applied to that one gene's p). - If signals are dense, TFisher with a high $τ$ approximates Fisher (including nearly all genes).

The "soft" weighting presumably smooths out the threshold effect so that one doesn't have to commit to a single $τ$. Possibly the CATFISH authors set a specific $τ$ (like 0.1 or 0.05) and a weight function. Without specifics, we assume soft TFisher is capturing *mid-sparsity scenarios that Fisher and minP aren't optimal for*.

Overlap with other tests: soft TFisher likely correlates with ACAT. ACAT itself is a form of weighted p-value combination: $\text{ACAT} = \tan\left(\frac{\pi}{2}(1-p)\right)$ sum. For moderate p (not extremely small), $ \tan(\frac{\pi}{2}(1-p)) \approx \cot(\frac{\pi}{2}p)$ grows like $1/p$ as $p \to 0$. That means ACAT effectively gives weight $\sim 1/p_i$ to small p's. This is a sort of smooth truncation: very small p's get huge weight, moderate p's get modest weight, large p's (close to 1) contribute almost nothing (since $ \cot(\approx \pi/2*0.5) \approx 0$). In effect, ACAT* continuously down-weights higher p-values *similar to a soft threshold. Soft TFisher might do something conceptually similar (like weight by $-\ln(p)$ up to a cap or so). So there is potential redundancy: ACAT and soft TFisher likely target a similar intermediate regime** where a few p's are very low and others are not, but not so extreme that minP alone suffices. We should examine correlation: - If a pathway has, say, 3 genes at p=1e-3 and 50 genes at p ~0.5, soft TFisher with threshold 0.05 would combine the three 0.001's strongly and ignore the 0.5's mostly. ACAT would also heavily weight those 0.001's (since $1/0.001 = 1000$ weight each, whereas 0.5 yields weight $\approx \cot(π/4)=1$ negligible). Both would likely yield significance. Fisher would also find it significant but with more noise from the 50 nulls (though 3 out of 53 p=0.001 might still show up significant in Fisher). - If signals are more widespread (like 15 genes at p=0.01 out of 100), Fisher and Stouffer will both detect it strongly, ACAT will too (each 0.01 gives $\cot(0.005π) ≈ 63$, summing 15 yields a large stat), soft TFisher presumably as well if threshold above 0.01 includes them. - If one gene is $1e-8$ and rest are null, soft TFisher with a low threshold (say 0.001) might consider only that gene (since others don't pass threshold), so then it behaves like minP or truncated combining of one value. ACAT will treat that 1e-8 as huge weight and others as ~0, so ACAT p will be about the same order as 1e-8.

So indeed ACAT and TFisher are both *tail-focused combination tests*, differing in the exact weighting function. It would be surprising if both are needed unless their behavior diverges in certain edge cases. Possibly "soft TFisher" is a custom approach by authors to ensure sensitivity to a middle fraction of significant genes that ACAT or minP might not fully exploit.

- **ACAT (Aggregated Cauchy)**: Already discussed earlier, ACAT combines p's via a Cauchy sum and is known to maintain validity under dependency and to *excel when one or a few p-values are extremely significant or a mix of some extremely and some moderately significant* [1]. ACAT's power profile often closely tracks the *best of Fisher and minP* in many scenarios. It was shown to be very robust and often as powerful as the optimal test in various rare-variant contexts [1]. ACAT has a heavy-tail property: one tiny p can dominate the sum (since $\tan((0.5-p)\pi)$ grows ~ $1/p$ as p→0, slower than log but still unbounded). Yet if there are many moderately small p, they add up too. Because ACAT does not require choosing a threshold, it's adaptive in a similar spirit to harmonic mean p or aSPU tests – it *automatically emphasizes the smallest p-values without discarding the rest entirely*. In many ways, ACAT might render one or more other tests redundant:
- ACAT has been touted as an omnibus for combining burden and SKAT in rare variants, proving robust across sparse vs dense signals [1]. By analogy, ACAT combining gene p's is likely robust across a range of pathway architectures.

- If ACAT performs ideally, one might argue it could replace the need for both minP and Fisher. In a sense, ACAT-O is already the *omnibus across different tests*, but here ACAT is one of the individual tests at gene-set level.

Why keep ACAT and minP separate if ACAT nearly encapsulates minP? Possibly because: - ACAT, while robust, might not capture the extreme-case as well as actual minP in small samples. For example, if one gene p=1e-6 and next best is 0.1, ACAT will yield a combined p maybe around 1e-6 or a bit larger (since the tan transform of 0.1 still adds some moderate positive value). minP would be exactly 1e-6 (calibration aside). So minP is a bit more *direct* for single-gene detection, whereas ACAT always leaves room that maybe multiple values contributed. - ACAT's validity under dependency is good, but the distribution used (Cauchy) is exact only if p's are independent Uniform(0,1) or certain dependency conditions. Simulations should calibrate it though.

ACAT's inclusion is certainly warranted as it's state-of-the-art for combination. It probably correlates with soft TFisher as reasoned, but ACAT might be more tail-sensitive (it's ~1/p for tiny p, whereas Fisher is ~log(1/p) and truncated might be fixed weight).

- **minP (minimum p-value)**: This test takes the smallest gene p in the pathway and uses that (with proper adjustment for gene count and correlation) as the pathway statistic. It is *most powerful in ultra-sparse scenarios – when only one gene (or a very small subset) in the pathway is truly associated*. It basically asks "is there at least one gene that is significant beyond what would be expected by chance?" The null distribution of minP is heavily influenced by m (number of genes) and their correlation, which CATFISH handles via MVN simulation or Brown's correction. minP has a strong track record: it's the basis of classical set tests like Tippett's test or the top-SNP in a gene set approach (GATES is a variant at SNP-level that picks best SNP with a Sidak correction). It will flag a pathway if *any single gene is extraordinarily significant* – e.g., if a pathway contains a known GWAS locus gene, minP will catch it (and yield a tiny p).

The downside is that minP ignores any incremental contributions from other genes. If a pathway has 5 genes with p~1e-3, minP is 1e-3 (not impressive after multiple testing across many genes perhaps), whereas Fisher or ACAT would have combined them to maybe $1e-7$. So minP is *underpowered for polygenic pathway effects*. It's likely included in CATFISH because it complements the others: it ensures that if there's a "star gene" in the pathway, the pathway gets credit for it. This is important because sometimes a single gene can drive a disease mechanism (like **FGFR2 in many breast cancer pathways** [38] [42]) – minP will make all those pathways look highly significant (for better or worse, as discussed).

Overlap & potential redundancy: ACAT and minP are highly correlated when one gene is much more significant than the rest. In that case, ACAT's statistic will be dominated by that gene's p-value (the term for that gene will be huge, others negligible). So ACAT p $\approx$ minP in extreme sparse cases. If multiple genes are very significant, minP only captures the smallest, while ACAT would benefit from also adding the second smallest, etc. So ACAT is typically more powerful if >1 gene is non-null; minP can be a bit less powerful unless that second gene's p is not much larger than the minimum.

Having both ensures that if a scenario arose where ACAT might be slightly suboptimal (perhaps correlation structures or moderate second signals), minP is explicitly tested. However, given ACAT's strong performance, one might think minP isn't strictly needed except for interpretability (minP pathways often indicate "this pathway is basically significant because of gene X").

Considering additional tests:

- **GSEA-KS (Gene Set Enrichment Analysis, Kolmogorov–Smirnov)**: This method (Mootha et al. 2003) ranks all genes by some statistic (e.g., gene-level Z or –log(p)) and then uses a running-sum (Kolmogorov–Smirnov-like) statistic to see if pathway genes are skewed toward the top of the ranking. It's a "competitive" test typically, meaning it compares pathway genes vs others. In a self-contained sense, one could implement a KS test: *are the p-values of genes in the pathway stochastically smaller than Uniform(0,1)?* The KS statistic would be $\max_{0<x<1} | \widehat{F}_{\text{genes}}(x) - x|$. Equivalently, one form of weighted KS is used in original GSEA.

Would GSEA-KS add something beyond the existing tests? Possibly: - KS is sensitive to *any consistent shift in the distribution of p-values*. It doesn't require specifying how many genes are significant; it looks for an enrichment of small p's without aggregating them in a specific way. In some sense, it's like a superset of tests that check k smallest p's for each k. In fact, the **Berk–Jones statistic** is quite similar to a KS-like approach but in a more calibrated way [10]. - If a moderate fraction of genes are somewhat significant, a KS test will pick up a leftward shift in the ECDF of p's. However, Fisher/Stouffer already do something similar by summing or averaging, which under many circumstances is as or more powerful (because sum is the optimal test for a mean shift, which is often how enrichment manifests). - GSEA's advantage historically was handling correlation via permutation and allowing weights (like weighting by gene-level effect sizes). But CATFISH is already directly accounting for correlation and using effect magnitudes inherently via p's.

One scenario: **if a pathway has an *unusual distribution* of p-values – e.g., many moderately low p (0.1) which wouldn't trigger Fisher or Stouffer as strongly, but more than expected by chance** – a KS test might detect that subtle shift. However, if p ~0.1, even KS won't be significant unless a large fraction are below 0.1 consistently. At that point, Fisher/Stouffer might also catch it if enough genes. Another scenario: KS might handle outliers differently – it's not as dominated by one small p (it's sensitive to the largest deviation in CDF which could be at a relatively high rank if one outlier at top doesn't drastically change the cumulative proportion).

Permutation reliance: GSEA-type tests often require permutation to get p-values because the distribution under null can be complex especially if gene scores are correlated. But in CATFISH's framework, we could calibrate a KS statistic similarly by simulation. However, since we already simulate for those five, adding a KS might be redundant with something like Berk–Jones (which is essentially a more powerful variant of a KS family test).

Conclusion: *Adding a GSEA/KS test likely would not substantially improve power given Fisher/Stouffer cover broad shifts, and ACAT/minP cover tails.* In fact, comparative studies (like one by Wang et al. 2010 or others) often found that parametric combination tests like Fisher have similar or better power than KS-based GSEA in many settings [43] [44]. Pascal specifically compared their continuous methods to a rank-sum (Wilcoxon) test, which is akin to a different non-parametric approach, and found continuous methods had *higher sensitivity and specificity* [43] [44]. They mention Pascal outperformed hypergeometric and rank-sum for detecting pathways in smaller GWAS data [44]. GSEA-KS is somewhat like a weighted version of these rank methods.

- **Berk–Jones (BJ) and Higher Criticism (HC):** These are *goodness-of-fit tests focusing on tail enrichment*. They look at $\max_{k} {\text{deviation of } \frac{#{p_i < t_k}}{\text{m}} \text{ from what's expected

under }U(0,1)}$ for some sequence t_k. BJ uses a range of t and effectively picks the one that maximizes a likelihood ratio; HC is similar but uses a specific normalization.
- **Generalized BJ (GBJ)** as used by Sun et al. is an analytic version that handles gene correlation and yields a p-value [11] [21]. They found GBJ more robust/powerful than HC in gene sets [11] [12].
- BJ/HC are known to be *asymptotically optimal for detecting sparse mixtures* in some regimes (HC for very sparse, BJ for moderate sparse).

In practice, BJ might do well if, say, a pathway has a modest fraction of genes with p way beyond uniform expectation. It in fact **includes minP and counts of top k as special cases in its scan**. Soft TFisher is like choosing a particular $t=τ$. BJ scans over all t adaptively. Therefore, **BJ could potentially be more powerful than soft TFisher if the optimal threshold isn't known a priori**. Soft TFisher picks one threshold (or a weighted scheme that implicitly has a threshold band). BJ will effectively consider all possible thresholds and take the best. The cost is that one then needs the null distribution of that maximum (which GBJ provides).

Would adding BJ meaningfully improve power? Possibly in niche cases where the distribution of gene p in the pathway is unusual (neither extremely sparse nor very dense, but something like, say, 5-10% of genes have p ~1e-3, others null). If soft TFisher's threshold was not well tuned to 1e-3, it might underperform, whereas BJ would find the best cutoff (like maybe $t=0.005$ yields maximal deviation). ACAT in many cases does a similar adaptive weighting, but not exactly a max over fractions. ACAT is like a weighted sum which is not identical to a max detection. There are scenarios where GBJ was shown to beat ACAT – for example, if there is a moderate proportion of true signals, ACAT still weights by 1/p which might overweight the lowest ones too much relative to a test that optimizes the fraction. However, ACAT is very competitive.

One advantage of including BJ is that it has **analytic calibration** (so adding it wouldn't require heavy simulation beyond what's done, if one uses their formula). If CATFISH team didn't implement it initially, adding it now could strengthen the method's scope, but with the simulation approach they can calibrate any test so that's fine.

So yes, **including a BJ test could capture scenarios that might slip through the cracks** of the current five. Given that soft TFisher is already an attempt in that direction, the incremental benefit might not be huge. But if soft TFisher uses a fixed truncation (maybe at 0.05 or 0.1), it may not always be optimal. BJ effectively gives an *omnibus over all possible truncation thresholds* [10]. This suggests:

> If possible, replacing or supplementing soft TFisher with a BJ-type statistic (or adding another test like GBJ) could improve power in intermediate cases. Sun et al. note that GBJ outperformed GHC (HC's analog) and was powerful in practical settings [11] [12]. It also protected type I error well [21].

However, adding BJ would increase correlation with soft TFisher/ACAT (since they all target tail enrichment). It might be better as a substitute: either use TFisher or BJ. BJ/HC are somewhat advanced; many general users might not be familiar with them, whereas Fisher, etc., are standard. Since CATFISH already deals with heavy calibration, it's feasible, but it complicates explanation.

- **Higher Criticism (HC/GHC):** This is extremely sparse oriented – HC looks at the smallest few p's but in a normalized way. In practice, for gene sets, GBJ was found superior to HC [11] [12], implying that if one were to choose, BJ is better for most realistic sets (not extremely sparse). If we had a scenario where *only a very tiny fraction of genes have weak signals* (like 1% of genes with p ~0.01 which might be too weak for minP but still a departure from null), HC might detect it. But in gene set sizes (~10–200 genes typically), 1% means maybe 0–2 genes. If 1% of 100 genes have p ~0.01, that's 1 gene with p=0.01 – minP would catch it actually with p=0.01. If 1% of 1000 genes in a huge set have p ~0.01 (10 genes), possibly none individually is <1e-4 but collectively it's something. But such large sets are unusual (most curated pathways aren't that large).

So HC might be more relevant in extremely high-dimensional contexts, not typical pathway sizes. GHC (generalized HC) exists, but given GBJ outshone it [11], focusing on BJ is better.

Overlap among existing five tests (especially soft TFisher):
We've partially covered this: - *Fisher vs Stouffer:* Both measure overall enrichment, with Fisher more tail-heavy. They will often produce correlated results. Soft TFisher is like a moderated Fisher. It's likely that Fisher and soft TFisher will often agree on whether a pathway is significant or not (soft TFisher might just be a bit more powerful in some middle cases). In fact, soft TFisher may make Fisher somewhat redundant unless the threshold is poorly chosen. Conversely, if threshold is high, soft TFisher approximates Fisher; if threshold is low, it approximates focusing on a subset like minP. - *ACAT vs soft TFisher:* as reasoned, ACAT's weighting ($\sim 1/p$) and a truncated Fisher ($-2\ln(p)$ for $p<τ$) are both emphasizing small p's. They differ in weighting scheme. Soft TFisher might be less extreme for ultra-small p's if threshold isn't too low (it might effectively cap contribution). ACAT has no cap; one extremely small p can produce an arbitrarily large statistic. So ACAT is more aggressive in tail emphasis. If a pathway truly has one tiny p, ACAT yields a more extreme combined p than a truncated Fisher that ignores everything else (depending on threshold). Actually, if threshold < that tiny p, truncated also uses it fully. But if threshold is moderate, truncated might include some nulls and thus have distribution with heavier df.

It would be useful to check if in practice ACAT and soft TFisher p-values are correlated. If yes, one might drop one. Or maybe they found that *soft TFisher covers some cases ACAT doesn't.* Perhaps if there are *multiple moderately significant genes but none extremely low*, ACAT will add them but might not produce as low p as Fisher or soft TFisher in that scenario (since ACAT weights each only by 1/0.01 ~100 for p=0.01, five such gives 500, $\tan^{-1}(500)$ yields a p still very small but depends). Actually ACAT would also yield a minuscule p if, say, five genes have p=0.01 – it would be significant but maybe not as strong as Fisher's sum of logs. Soft TFisher might shine if exactly, say, half the genes have p < 0.05, because it ignores the half that are >0.05 and sums logs of the significant ones, boosting signal. ACAT in that case uses all of them but high p's contribute ~0 anyway, so it effectively uses those <0.05 with weight ~some value. Possibly similar outcome.

- *minP vs ACAT:* as said, minP is contained in ACAT's behavior for one extreme p. Redundancy is high in one-hit cases. But minP ensures direct calibration for the one gene scenario, which ACAT might approximate but maybe not exactly match distribution-wise.

Given these overlaps, one might consider trimming the test battery. For example: - If one added **GBJ**, one might drop **soft TFisher**, as GBJ would cover the truncated approach in a more optimal way. - Or drop Stouffer or Fisher if one feels they add little distinct info. Many pipelines pick either the sum of logs or sum of z, not both. Since CATFISH can calibrate anything by simulation, it's not a problem to include redundant ones aside from slight efficiency loss and correlation among tests (which might slightly reduce the benefit of having multiple because combining correlated tests yields diminishing returns).

But perhaps the authors kept all five to demonstrate thoroughness and to feed the omnibus diverse inputs. The omnibus ACAT-O or minP-O will naturally downplay redundant tests because if two tests always produce similar p's, they won't add independent evidence – ACAT combining them is almost like counting the same signal twice, but since they are correlated, the omnibus calibration accounts for that.

Directional Stouffer's test (one-sided combination):
As touched earlier, a directional test would combine signed Z-scores: - Define for each gene a Z (positive if risk allele increases trait, negative if decreases). Then use Stouffer's method: $Z_{\text{dir}} = \sum_{i} w_i Z_i / \sqrt{\sum w_i^2}$. This essentially tests H_0: "the gene effects have mean zero (no net effect in either direction)" vs H_a: "the pathway's genes tend to have effects in one direction (positive or negative)". It is a *different alternative* than the typical two-sided enrichment. - If a pathway truly influences trait in a concerted way (all genes in pathway increase disease risk when mutated, for example), a directional test is more powerful because it leverages that consistency. A two-sided test like Fisher or normal Stouffer would treat a protective gene (negative Z) as just another small p (since two-sided p might be small for a big negative Z too). That still contributes to significance in a non-directional way, but it doesn't distinguish sign. - If a pathway has mixed directions, a directional test could cancel out signals. For instance, one gene raises risk (Z=4), another lowers risk (Z=-4). A non-directional Fisher sees both p ~ 3e-5, very significant enrichment collectively; a directional sum Z would sum to 0, indicating no net effect. So they test different hypotheses (self-contained vs competitive directional).

When would directional be used? Possibly in settings like gene expression pathways or when one suspects a pathway's activation (or suppression) is the relevant phenomenon. In GWAS, pathways are not so straightforward – genes can have risk or protective alleles. Unless we have some prior that "in this pathway, if variants disrupt it they all cause disease in same direction", which is not guaranteed, a directional test could miss actual associations. Typically, gene-set analyses in GWAS ignore direction (since effect allele assignment is arbitrary).

However, consider polygenic score type pathways: if genes were scored by effect on trait, one might check if they are mostly positive. Some tools used a *signed sum of z* for pathways (e.g., some pathway scoring in MAGMA can include direction if doing regression with gene effect signs – but usually not).

A suggestion might be: after identifying a pathway, check if the associated genes all have effects in the same direction (which could lend mechanistic insight).

Possibly, including a directional test as an additional statistic in CATFISH could allow detecting pathways where genes act consistently. But if a pathway has both protective and deleterious members, the non-directional tests are more appropriate. Including directional tests would likely only help if such consistent pathways exist and were missed by the others. They might – e.g., in metabolic pathways, maybe all genes that increase metabolic rate lower obesity risk, giving consistent directions.

Implementation: One would need to pick a sign for each gene. If using MAGMA p's (which are two-sided), one can get a direction by looking at the top SNP effect or performing MAGMA one-sided tests (like testing positive association separately from negative). Possibly the authors did not incorporate this because it complicates the pipeline (one needs effect size info, not just p's, and a decision how to combine).

If it were included, you'd have two tests per pathway (one for positive, one for negative direction). Then the pathway's "directional p" is the smaller of those two (or you could incorporate it in the omnibus by just including both as separate tests).

Considering correlation, a positive-direction test and a negative-direction test on the same data will be highly negatively correlated (if one is significant positive, it won't be negative significant). This could slightly complicate the omnibus calibration but not too much if accounted.

Assessment: Using directional Stouffer is an interesting idea but likely *not necessary for general power*. It addresses a specific question: *does the pathway show unidirectional association?* In most GWAS, we just want any association regardless of sign, which is captured by two-sided tests. Unless prior knowledge strongly suggests directional consistency, adding it might add complexity without broad benefit. It could even be detrimental: if an important pathway has 3 risk-increasing and 2 risk-decreasing genes, non-directional tests catch it (they see 5 small p's), but directional splits them and maybe neither direction passes threshold.

So, perhaps directional tests are best left as a post-hoc analysis of significant pathways rather than part of the discovery procedure.

Summary of test choices critique:

The five tests in CATFISH cover a spectrum: - Fisher/Stouffer: whole-distribution shifts (dense polygenic effects). - soft TFisher: mid-range sparsity (some fraction of genes are effect). - ACAT: adaptive weighting, good for sparse or mix. - minP: single-gene detection (ultra-sparse).

This is a reasonable set and likely ensures the omnibus has good sensitivity across scenarios. The **overlap** between these methods is significant – e.g., ACAT and soft TFisher both emphasize small p's; Fisher and Stouffer both emphasize overall shift. But by including all, CATFISH's omnibus can leverage whichever yields the smallest p for a given pathway.

If one were to streamline: potentially drop one of Fisher/Stouffer (since their performance is usually similar), and possibly replace soft TFisher with a more formal adaptive test like BJ. But even as is, the redundancy is not a huge issue except that it might not increase power as much as adding a truly orthogonal test would.

Could adding GSEA-KS, BJ, or HC improve power?

- *GSEA-KS:* Unlikely to add beyond what Fisher/Stouffer do, especially since those have been shown to outperform or match KS in many settings [43]. It also complicates calibration and interpretation (the KS statistic isn't as straightforward to explain).
- *Berk–Jones (GBJ):* Yes, this could improve detection for intermediate scenarios. If CATFISH found that soft TFisher's threshold was suboptimal in some sims, GBJ could solve that by not requiring a fixed threshold. If resources allow, including GBJ would strengthen the method's pedigree (it's a known optimal test for set association in some stat literature [10] [12]). On the flip side, ACAT already mimics an adaptive test and might capture much of that benefit. But I suspect there are cases where GBJ would outperform ACAT or soft TFisher. For instance, if 10% of genes have p ~1e-3 and others are null, ACAT would sum ~1000 for each of those 10, total ~10000, giving a very low p, actually that might be fine. If 20% genes at 0.01, ACAT: each ~$\cot(0.0157) ~ 20$, sum 0.2*m*20, so e.g. for m=100,

20*20=400, ACAT stat leads to p extremely small too. Hmm, ACAT actually does well there. Perhaps differences are subtle. Regardless, GBJ would ensure that scenario is caught.

- *Higher Criticism:* Perhaps not needed separately if we have minP for ultra-sparse and ACAT/GBJ for moderate. HC's regime (very faint but very sparse signals) might not realistically appear given the discrete nature of pathways (if only 1 or 2 gene have weak effect, minP or Fisher likely sees them if sample size is large enough for them to pass nominal significance at all).
- *Directional Stouffer:* Only useful if expecting coherent direction. For complex traits, we usually don't impose that assumption at discovery stage. So, it's not recommended as a primary test. It could be mentioned as a future extension: e.g., "If prior evidence suggests all genes in a set have effects in the same direction, a one-sided Stouffer test could boost power by accounting for sign, but we did not use this due to risk of missing pathways with mixed effect directions."

Conclusion for Section 4:
The five tests chosen are generally sensible and cover complementary alternatives. Soft TFisher's role is to handle "some genes significant, others not" scenarios; ACAT covers a wide range effectively; minP catches the single gene spikes; Fisher/Stouffer cover bulk shifts. Adding tests like BJ (Generalized Berk–Jones) *could* further improve power for intermediate proportions of associated genes, as it is known to be an asymptotically optimal test for certain alternatives [10] [12]. If computational resources and complexity aren't a concern, including GBJ might be worth it – it would basically subsume what soft TFisher is trying to do with a fixed threshold by doing it optimally and analytically.

Otherwise, the current set is adequate. Redundancy in test choices mainly means the omnibus is slightly less efficient (because it's combining correlated tests), but since CATFISH calibrates the dependence, the worst case is a small loss of power due to "multiple testing penalty" for including similar tests. This might be minimal if tests are highly correlated (because effectively you're not gaining independent information from each). If one were refining the method, one might prune one of a pair of highly correlated tests to reduce that penalty. For instance, if analysis shows Fisher and Stouffer nearly always produce similar p's, one could drop one to reduce the dimension of the omnibus combination (makes simulation a tad simpler too).

In summary, the five tests chosen form a reasonable basis, and *the addition of another tail-focused test like GBJ could be considered to strengthen performance for moderate-sparsity signals*. Directional testing is probably not broadly useful except in specialized scenarios, so we don't recommend it as a default component of CATFISH (it could miss pathways where genes have opposite effects, which is quite possible biologically). If a specific hypothesis about direction exists, one could apply a directional test separately.

5. Gene-Level Modeling in MAGMA and Alternatives

CATFISH uses **MAGMA** gene-level p-values as input, so the quality and characteristics of those gene scores are critical. We assess MAGMA's sensitivity to its parameters (window size, model, LD reference) and compare it to other gene-level association methods like Pascal, VEGAS2, fastBAT, and SKAT-based tests. We also suggest how to benchmark the gene-level approach to ensure CATFISH is built on a solid foundation.

MAGMA gene-level analysis:
MAGMA (Multi-marker Analysis of GenoMic Annotation) is a popular tool that performs gene-based

association using a multiple linear regression model of SNPs on phenotype [25]. Key points about MAGMA: - It aggregates SNP data within a gene (usually including SNPs in a specified window around the gene) and computes a test statistic (typically an F-test) for the joint effect of those SNPs [25]. This accounts for linkage disequilibrium (LD) by modeling the correlation structure of SNPs – effectively, it's like a principal components or regression approach that keeps degrees of freedom in check. - MAGMA can output a gene p-value that is properly calibrated even when SNPs are in LD, given a reference panel for LD. It also corrects for gene size and SNP density to some extent by the regression framework (i.e., it's not simply summing statistics, it's testing if any linear combination of SNPs has effect). - By default, MAGMA typically uses a **gene window of ±50 kb** around the gene boundaries (although users can choose differently). This is to capture regulatory variants just outside coding regions. The choice of window can affect which variants are assigned to a gene: a larger window might include relevant regulatory SNPs (improving gene signal if those SNPs are truly affecting gene function) but also can include noise (SNPs that might actually affect a neighboring gene or be non-functional). It can also introduce correlation between gene tests if windows overlap (adjacent genes might share SNPs).

Sensitivity to window size: If the window is too small (e.g., only exonic SNPs), one might miss associations from regulatory variants (e.g., an enhancer 100 kb away that loops to the gene). If too large, one risks diluting the association or attributing signals to a gene that truly belong to a different gene's regulatory element. MAGMA's default 50 kb is a compromise from literature (it often captures most promoter regions). Some analyses use ±100 kb for completeness, at risk of more overlaps. There is also **H-MAGMA** which uses Hi-C data to assign non-linear regulatory connections – not just distance [45] [46]. - It might be worth exploring if *H-MAGMA annotations* (which assign SNPs to genes based on 3D contacts and eQTL data) could improve gene p-values [45] [46]. If CATFISH's user or authors have access, H-MAGMA can reassign distal SNPs to the appropriate gene, potentially increasing gene signal for genes influenced by distant variants (common in brain and other tissues). The trade-off is complexity and possibly adding false positives if contacts are not relevant.

For now, assuming standard MAGMA: - A **smaller window** (e.g., ±20 kb) would result in more conservative gene definitions with less overlap. It might reduce gene–gene correlation (which simplifies pathway analysis calibration), but risk missing signals from enhancers further away. - A **larger window** (like ±100 kb) might slightly inflate type I error at gene level if not accounted (due to including noise SNPs or double-counting between neighbors). MAGMA can handle correlated SNPs fine, but if two adjacent genes share a lot of SNPs, their gene tests become correlated (which we do handle later in pathway calibration though).

It's good to examine if results are robust to reasonable window changes. Perhaps do a sensitivity analysis: e.g., run MAGMA with 20 kb, 50 kb, 100 kb windows, see if key gene findings (and subsequently pathway findings) change drastically or if calibration of pathways worsens (excess correlation from big windows can create more pathway null inflation if not fused).

MAGMA model choices:
MAGMA's gene test by default is a *"snp-wise mean model"*, which basically tests if the mean effect of SNPs in the gene is non-zero, controlling for LD [25]. This is akin to a multi-SNP regression. There is also a *"snp-wise top model"* which tests if any SNP in the gene is significant (similar to minP at gene level). MAGMA, I believe, allows an option to use the top SNP's p-value but correct for gene size (like a Sidak or GATES method). - Using the *mean model vs top model* can yield different gene p's. The mean model (which sums contributions) is more powerful for genes where multiple SNPs each have moderate effect (polygenic within gene, or multiple independent hits in a gene). The top SNP model is more powerful if only one variant in the gene is

truly causal and others are noise – it effectively picks the min p in that gene (like a burden test vs a min test). - MAGMA's default is the mean model (multi-SNP) because it's more general and tends to outperform simple top SNP especially for common trait architectures where a gene might have several contributing variants of small effect. - However, if a gene has one very strong SNP (like APOE gene with rs7412 for Alzheimer's), the mean model will detect it, but the top SNP model would too and maybe with similar p (since that SNP dominates). - One could consider an *adaptive gene test* combining both (like SKAT-O does with burden vs SKAT). MAGMA doesn't directly do an adapt, but one could take min(gene p from multi-SNP, gene p from top-SNP) as a gene-level stat. But that then needs calibration. - Alternatively, using other tools like **Pascal**: Pascal offered *sum of chi-square* (akin to an unweighted burden which is similar to MAGMA's multi-SNP test) and *max of chi-square* (like top SNP) gene scores [47] [48]. They found both had merits; they defaulted to sum for pathway analysis but also looked at max for certain detection [49].

LD reference sensitivity:
MAGMA requires an LD reference panel to estimate SNP correlations. Commonly used references are 1000 Genomes or UK Biobank. - If the reference doesn't match the study population, gene p-values can be off. For instance, using European LD for an East Asian GWAS might mis-estimate gene variance (SNPs in high LD in EA but not in EUR could lead to underestimation of effective number of SNPs, making gene test slightly anti-conservative or vice versa). - Within European populations, smaller reference sets (e.g., 1000G EUR with ~500 samples) might introduce noise in correlation estimates, though MAGMA usually handles it. Pascal's authors compared references and found stable results in a large HDL GWAS when using an internal vs 1000G reference.

Still, for precision: If the GWAS sample is huge (say UK Biobank with N~300k), using a *subset of that data as LD reference* is ideal (to capture fine-grained LD). If only summary stats are available, 1000 Genomes or HapMap are often used. MAGMA can also use the *"--effective n" and "--window" flags* to adjust. - The effect of reference mismatch might be small on final pathway results, but it could matter for borderline gene p's. - We recommend using as high-quality an LD panel as possible (matching ancestry and large sample). For example, if UK Biobank is used, perhaps use UKB controls or HRC reference with thousands of samples to compute LD. - Another parameter: **Gene covariates**. MAGMA's gene-set analysis (competitive test) often includes gene length, gene density as covariates to avoid bias that large genes (with more SNPs) have smaller p by chance [26] [30]. In CATFISH's self-contained approach, this is not directly used, but one might still observe that larger genes can have smaller p more often purely due to containing more SNPs (more "shots" to find a significant SNP). MAGMA's regression approach mitigates this by properly accounting degrees of freedom. Indeed, Pascal observed that without correction, pathway p-values could be inflated if many large/LD-correlated genes are in a pathway [26] [50]. They solved it by gene-fusion for LD and by not needing explicit length covariate because their method inherently normalizes (they did mention hypergeometric is biased by pathway size, etc., but continuous methods less so).

Comparing MAGMA to other gene-level methods:

- **Pascal (2016)**: Pascal uses summary stats to compute gene scores either by *sum of chi-square* (like a sum of χ^2 of SNP p's, which is akin to a fixed-effect test capturing cumulative effect) or *max of chi-square* (the single strongest SNP). It uses an analytic approximation (Chi-square mixture or numerical integration) to compute gene p, avoiding heavy simulation [51] [52]. They took LD from 1000 Genomes and even fused neighboring genes if needed [26] [27]. They claimed rigorous type I control and more power than VEGAS/PLINK in benchmarks [53] [54]. Pascal also used a "modified

Fisher method" for pathways with empirical adjustment [49] [55] – we covered that earlier. In comparisons:

- Pascal vs MAGMA: Both aim to do similar things. MAGMA's regression vs Pascal's chi-square sum are conceptually similar for gene-level if trait is continuous. For case-control, MAGMA uses logistic regression approximations, Pascal uses χ^2 from association test.
- One difference: MAGMA can incorporate sample size or case-control ratio properly (regression knows variance), whereas Pascal just uses summary stats and might have to approximate correlation of test stats. But they did well, verifying results matched VEGAS high precision [56].
- **Benchmarking**: One could run Pascal on the same summary stats and compare gene p-values with MAGMA's. Pascal's authors did analogous comparisons with VEGAS and found high concordance except at extremely low p where Monte Carlo resolution limited VEGAS [57]. If MAGMA and Pascal give very different gene p for a gene, that might indicate an issue (e.g., MAGMA might spread effect across correlated SNPs differently).

- If one is worried about MAGMA missing signals due to multi-collinearity, one could combine approaches: e.g., keep MAGMA but also calculate a gene p by a simpler method (like pick top SNP p as gene p). If a gene has a very significant SNP but MAGMA's multi-SNP p turned out not so low (maybe because adding other SNP covariates reduced significance?), that could be a sign. Usually, though, if one SNP is that strong, MAGMA's F-test will still be significant (the SNP's effect can't be fully "cancelled" by adding others, they would just soak up residual df).

- **VEGAS/VEGAS2 (Liu et al. 2010, 2013)**: VEGAS uses a simulation or numerical integration of the sum of χ^2 method (like akin to Pascal's sum but via simulation). It was one of the first widely used gene tests for GWAS. VEGAS2 extended to allow larger gene sizes by precomputing LD blocks.

- Performance: Pascal showed that analytic calculation can replace simulation efficiently with identical results [56].
- Power: Not significantly different from MAGMA or Pascal if done equivalently.
- A difference: VEGAS originally did not explicitly account for multiple independent signals unless one used a sliding window approach. Pascal's sum of chi-square effectively captures cumulative signals. MAGMA's regression also can capture multiple signals.
- So likely MAGMA, Pascal, VEGAS yield similar p's for genes with one moderate signal or multiple small.

- If a gene has multiple independent SNP associations, a regression test (MAGMA) can actually incorporate that better: because multiple independent SNPs will each contribute partial R^2, raising the F-stat. A simple sum of χ^2 might overweight correlation though – but they account for LD so it's fine. Actually, if multiple independent signals, both sum of chi^2 and regression should detect stronger association than just the top SNP.

- **fastBAT (Bakshi et al. 2016)**: fastBAT is another method that also uses an analytical approach to combine SNP p-values using LD information [58] [59]. It essentially does something similar to VEGAS but in a much faster matrix algebra way (exploiting correlation structure). They claimed it's extremely fast and as accurate as simulation-based. They also mentioned identifying novel genes not found by single SNP analysis in large meta-analyses [59] – specifically due to "multiple small independent signals at these loci" [60]. That underscores that gene-based tests (including MAGMA or fastBAT) can find genes where no single SNP reached genome-wide significance, but collectively the SNPs indicate association (e.g., FOXP1 in schizophrenia with multiple SNPs of moderate effect) [60].

- If using MAGMA, presumably it would also find those genes. It'd be interesting to cross-check: Did MAGMA highlight the same novel gene signals that fastBAT did? Probably yes if the data and approach are similar.
- The differences between fastBAT and MAGMA are subtle – both rely on LD info to combine SNP p's. Possibly fastBAT approximates a SPU (sum of powers) test with power=2 (like sum of chi2).
- Since CATFISH already uses MAGMA, switching to fastBAT likely wouldn't change much, but one could double-check known instances like the ones in fastBAT's paper [59]. If MAGMA missed any gene that fastBAT flagged, that would be concerning – but likely not.

- **SKAT and burden tests:** In a GWAS context, SKAT (Wu et al. 2011) is a sequence kernel association test often used for rare variants. But for common variants, SKAT can still be used to test a set (like all SNPs in a gene) by treating them as random effects. SKAT is essentially a variance-components test ~ sum of (w_i * beta_i)^2 test, which for common variants ends up similar to linear combination of SNP chisq. A burden test is sum of SNP effects (good if effects are in same direction).

- For common variants, and under an additive model, a linear regression (like MAGMA's) is equivalent to a multivariate Wald test on coefficients, which is akin to SKAT's approach if all SNP effects are considered fixed. SKAT as a score test in a random effects model might differ if effect signs vary, but multiple regression captures that by giving each SNP its own coefficient (like SKAT).
- Actually, in rare variant tests, SKAT vs burden differences matter because signs and effect directions vary. For common variants, we often assume each SNP can have independent effect (like MAGMA does).
- MAGMA's gene test is essentially the *fixed-effects analogue of SKAT-O*: It will detect association if either one SNP or multiple contribute, regardless of sign, because each SNP has its own param. A burden test would fail if SNP effects cancel out (some increase risk, some decrease risk).
- If we suspected a gene has variants with opposite effects (e.g., both loss-of-function protective and gain-of-function risk variants in same gene), a burden test would lose power whereas MAGMA (or SKAT) would still find gene association.
- So MAGMA is more robust in that sense than a simple burden. SKAT-O tries both burden and SKAT and adaptively combines – but for common variants typically SKAT (which allows different directions) is safer.
- There's also **KGG** software (Li et al.) which offered GATES (a top-SNP test) and HYST (hybrid set test) which merges top signals from different LD blocks within a gene – basically acknowledging gene might have multiple independent signals and trying to combine a few top SNPs rather than all. That's somewhat between minP and sum.
    - If MAGMA only did an omnibus test for any effect (which it does), it is akin to combining signals across blocks anyway (the regression will pick up multiple independent effects by higher test statistic). HYST tries a structured approach: pick best SNP from each of, say, two blocks, then combine those two p's by Fisher. This was shown to improve power when a gene has, say, two independent hits in different LD blocks [61]. But MAGMA's multi-SNP should also capture that as strong joint effect.

Benchmarking strategies:

To ensure the gene-level approach is solid, we propose: - **Simulation tests:** Simulate phenotypes with known gene-level architecture: 1. Choose a gene and simulate a phenotype effect for one SNP in that gene (to mimic a single-causal variant gene). See if MAGMA's gene p is significant as expected and compare to e.g. minSNP p. This checks MAGMA's calibration (should yield p ~ single SNP's p or slightly less extreme if adding other SNP covariates). 2. Simulate a gene with two causal SNPs (independent LD blocks). Does MAGMA pick it up better than a single SNP test would? It should, showing higher chi-square than either SNP alone. 3. Simulate many small effects across the gene's SNPs (polygenic within gene). MAGMA's regression should detect it if aggregated effect is enough, whereas a top SNP might not. Confirm that. - These simulation can use real genotype data for LD realism or artificial correlation structures.

- **Cross-method comparison on real data:** Run a few gene-based methods on the actual summary stats:
- MAGMA (current approach).
- Pascal's sum and max.
- Possibly SKAT common variant test (if individual genotypes are available or summary-level approximations can be used, though summary SKAT is tricky without full info).
- See how the gene p-values correlate. Ideally, significant genes should overlap. If MAGMA missed any gene that another method finds with high confidence (and we suspect it's a true gene), it could indicate MAGMA's defaults might not be optimal (like maybe top SNP method was needed).

- If differences are minor (just random variations for marginal genes), that's fine.

- **Check MAGMA's assumptions:** MAGMA uses a linear model which assumes additive effects and normal phenotypes or approximations for case-control. If the trait is case-control with strong effects, logistic might be more accurate. But MAGMA's documentation indicates it handles binary by a linear regression approx or by including case count information. For huge samples, difference is minimal.

- **Gene-level QC:** Sometimes gene p-values can be slightly anti-conservative if a gene has very few SNPs (like 1 or 2 SNP genes) or extremely high LD (making matrix inversion tricky).

- MAGMA likely applies a correction for genes with 1 SNP (then gene p = SNP p).
- Genes with many SNPs: It uses an approximation for degrees of freedom (like effective number of independent SNPs).
- The authors of MAGMA and Pascal both stress they maintain type I error for genes of varying SNP counts [35] [36], [53].
- In extremely gene-dense regions (MHC), gene tests can be complicated due to correlated signals across genes. That again enters pathway correlation if those genes in same pathway.

Alternate approach – conditional testing between genes: - When genes are very close (or overlapping windows), one could consider merging them as earlier or performing conditional analysis: e.g., if gene A and B overlap, test gene A after removing SNPs assigned to B etc. MAGMA does not do that by default, but gene fusion in Pascal effectively does by merging them into one locus. - If CATFISH finds pathways enriched purely because of a locus with multiple nearby genes (like HLA region where many adjacent genes show association due to LD with same causal variant), that could inflate pathway significance (multiple genes from one locus all low p). A pathway containing all those genes would seem to have many "hits" but really one locus. - Ideally, one could do a conditional MAGMA analysis: e.g., condition on top SNPs in region to see if multiple gene signals are independent or the same. But that's advanced and probably beyond scope. Instead, addressing it at pathway stage by flagging single-locus-driven pathways is approach (point 6).

Comparing speed and practicality: - MAGMA is very fast and handles genome-wide easily (minutes). Pascal and others also are fast nowadays (Pascal was also minutes on thousands of genes with precomputed mats). - One note: MAGMA's output includes a gene-gene correlation matrix for gene-set analysis if you ask for it, which CATFISH presumably uses (or calculates similarly). That's helpful. - Another note: If some gene has extremely high LD among its SNPs, MAGMA might drop one SNP or regularize. Not likely an issue unless using huge windows.

fastBAT, SKAT examples – The fastBAT paper's mention of novel genes means: - Without gene test, those genes wouldn't reach significance (no single SNP passed 5e-8), but gene p passed a threshold like 2.5e-6 (since they reported as novel at gene-level significance). - We should ensure CATFISH leverages such gene-level discoveries. If a pathway is enriched with genes that individually were just below SNP significance but gene tests raised them, that pathway might pop out even if no single SNP did. This is a major benefit of pathway analysis – capturing modest associations spread across multiple genes. - So MAGMA doing multi-SNP helps by bringing such genes to attention. If we had used a top SNP per gene approach for pathways (like just minP at gene level), we might miss polygenic genes. Good that MAGMA covers that.

In summary for Section 5:
MAGMA is a robust choice for gene-level p-values, offering an effective balance of power and calibration. Its sensitivity to window size should be examined – e.g., does extending windows or using functional annotations (H-MAGMA) appreciably change results? The default (50 kb) is generally reasonable [45] [46], but if a trait's associations are mostly within coding regions, smaller windows could reduce noise slightly. Conversely, if regulatory variants far from genes are known to be important (like enhancer variants), one might incorporate them via larger windows or via annotation-based mapping.

Comparing to Pascal/VEGAS/fastBAT suggests MAGMA's multi-SNP test is on par or better in power. Each method has slight differences but all incorporate LD. It's important to verify that MAGMA isn't systematically losing any signals: - A *possible edge case:* If a gene has two distinct association signals in complete LD blocks separated by, say, 1 Mb (so essentially it's a big gene or gene family region), MAGMA's regression includes all SNPs – it will still detect combined effect. Pascal's sum of chi2 does too. If signals are opposite in effect (rare in same gene for same trait), MAGMA would still give a strong p (because F-test doesn't care direction per SNP).

One thing to watch: If the LD reference isn't perfect, gene p-values could be slightly inflated or deflated. But since we calibrate pathway null by simulation ultimately, any mild mis-calibration at gene level might be mitigated at pathway level (the simulation draws from an estimated gene correlation that would reflect gene p uniform assumption – if gene p are slightly conservative, then under null maybe fewer pathways appear significant, which is fine).

Benchmark suggestion: Use known positive controls: - If known gene-level associations exist (like established GWAS genes), see that MAGMA yields very low p for them. If an alternative method yields even lower, check why (maybe MAGMA's considering correlated SNPs reduces significance slightly vs a single-SNP test). - The difference is often minor: e.g., if a gene has one SNP p=1e-10, MAGMA might output gene p ~1e-10 or sometimes slightly higher if it had to account 10 SNPs with that one being standout (the degrees of freedom means test is on 10 df instead of 1, but one df has huge F, so still extreme p but maybe 1e-9). - However, MAGMA's default uses an F-test with m SNP df in numerator – one extremely associated SNP will yield an F roughly = $\chi^2(1)$ with 1 df if others add nothing, and trivial contributions of others might degrade it a little (like p might become 5e-10 instead of 1e-10). In large data, that's negligible difference.

Gene-level alternatives and publication: - The authors might mention in discussion: "We chose MAGMA for gene analysis due to its efficiency and widespread use [36], but similar results could be obtained with other methods (e.g., Pascal, VEGAS2, fastBAT) [56] [59]. For thoroughness, one could benchmark CATFISH using gene p's from different methods; we anticipate results to be highly concordant, as all these methods properly account for LD in combining SNP signals." - It's reassuring to readers that the pathway findings are not an artifact of a particular gene scoring method. If a pathway is truly enriched, it should show up whether you use MAGMA's gene p or Pascal's, etc., aside from random variation. If not, then the method choice would be questioned.

So, to prepare CATFISH for publication: - Confirm gene-level analysis is robust by cross-checking or simulation. - Possibly mention if any limitations: e.g., "Genes with extremely high LD or gene clusters may challenge gene-level tests. We mitigated this by using high-quality LD reference and by merging highly overlapping genes (as discussed earlier) [26] [27]." - And "We compared our MAGMA-derived gene p-values with Pascal's method and found high concordance (Spearman 0.98 for gene p, data not shown), indicating our gene-level results are not sensitive to the choice of method." This kind of statement builds confidence.

In conclusion, MAGMA provides a solid baseline. By comparing with alternatives and maybe tweaking parameters, we ensure CATFISH's results are not overly driven by an arbitrary gene-level decision. One improvement for the future could be incorporating *tissue-specific mapping of SNPs to genes (H-MAGMA)* to capture non-coding signals better [45] [46] – that would assign SNPs to genes they regulate (maybe including non-neighboring genes). This might reveal pathway links that standard mapping misses (for example, an intergenic SNP affecting a pathway gene's expression in a certain cell type could be missed if it's >50kb away and not assigned). Including such advanced mapping is beyond standard practice, but could be mentioned as a potential extension for refining gene-level input.

6. Managing Pathway Redundancy and Overlap

Pathway databases are notoriously redundant – gene sets often overlap heavily or represent nested concepts. This presents challenges: - Multiple pathways may be "significant" due to the same underlying gene(s) (single-gene or single-locus effects). - Highly overlapping pathways (like several GO terms sharing most genes) can all light up together, giving a false impression of many findings, when it's essentially one biological signal. - Some "pathways" are so small or trivial (e.g., a set containing 1-3 genes, or a gene family that is essentially one gene with isoforms) that calling them enriched is not informative ("single-gene proxy pathways"). - Conversely, very large pathways can subsume smaller ones (if the big one is significant, some subcomponents might also appear nominally significant or vice versa).

Strategies to address these issues:

1. **Pre-filter or annotate pathways by size/content:**
   It's common to set a lower bound on pathway size – e.g., analyze pathways with at least, say, 5 or 10 genes to avoid tiny sets. Single-gene "pathways" are usually removed because they contribute nothing beyond the gene-level result. If not removed, they should be *flagged*:
2. We can automatically label any pathway where $m=1$ (or very small m) as essentially "single-gene". If that gene is genome-wide significant itself, then obviously the pathway will be too (minP = gene p) [38]. Reporting it as a pathway finding is redundant. Instead, the interpretation should be that the association is driven by that gene.

3. In a publication, one might explicitly state: “We excluded gene sets with <5 genes from enrichment analysis to focus on broader pathways.” If not excluded, at least in results one would note if a top pathway has only 1-2 genes, likely it’s not a real pathway effect but the individual gene effect.
4. Example: If a KEGG pathway only has gene *TP53*, and TP53 is significant, then obviously that pathway shows up. It’s not adding knowledge beyond “TP53 is associated.”

5. We can also examine **single-locus pathways**: e.g., FGFR2 is a gene; any pathway that includes FGFR2 might be significant if FGFR2 is strongly associated [38]. That doesn’t mean all those pathways (be it “FGFR signaling”, “Ear morphogenesis”, etc.) are independently implicated – they are proxies for that gene’s association [38] [42].

6. **Clustering or grouping similar pathways:**
   Use gene overlap or semantic similarity to cluster pathways into “themes”. For instance:

7. Build a graph where nodes are pathways and edges connect pathways with Jaccard overlap > some threshold (like >0.5). Significant pathways often form clusters in this graph (because if they share many genes and one set is significant, the other likely is too).
8. One can then compress results by representing each cluster by one pathway (perhaps the one with the highest significance or the one most interpretable).
9. Tools exist: e.g., *EnrichmentMap* (a Cytoscape plugin) which visualizes enriched GO terms as a network, merging ones with high overlap. *REVIGO* is another that reduces GO terms by semantic similarity.
10. For a publication, one could present clusters: “We identified 3 clusters of enriched pathways: (1) a cluster related to *T cell activation* (comprising GO terms X, Y, Z all containing many common genes), (2) a cluster related to *FGFR2-related pathways*, etc.” This provides a higher-level interpretation rather than listing 20 redundant terms.

11. At minimum, we can note in the text: “Many of the significant pathways are highly overlapping. For example, out of the top 10, 6 pathways contain a common core set of genes on chromosome 10 (driven by gene ABC). After conditioning on gene ABC, those pathways lose significance, suggesting they are not independent signals.” This is similar to what Sun et al. did with step-down analysis [62] [42].

12. **Conditional testing or step-down approach:**
    This is a more formal way to handle redundancy:

13. The idea (as in Sun et al. 2019) is to iteratively remove the contribution of top genes and re-test pathways [62] [63]. By doing a “step-down” procedure, one can distinguish pathways that still show enrichment after removing single-gene effects vs those entirely driven by one or two genes.
14. For example, FGFR2: They removed FGFR2 from all pathways and found that many pathways lost significance [38] [63], indicating those were FGFR2-driven signals. The remaining truly enriched pathways had more distributed signals.
15. CATFISH could implement a similar post-hoc: For each significant pathway, check if removing its most significant gene (set that gene’s p=1 or drop it and recalibrate the pathway p by simulation quickly) makes the pathway non-significant. If yes, then the pathway was “single-gene-driven”. One can then label it as such in results.
16. Or on a broader scale, identify if the same gene (or small set of genes) appears in many significant pathways. If yes, then those pathways are likely not independent findings.

17. A simpler approach: list the top contributing genes for each significant pathway (e.g., the genes with lowest p in that set). If you see the same gene in many lists, that gene is a driver. You might group those pathways under that gene's umbrella.
18. This is akin to *conditional enrichment analysis* used in gene expression: e.g., test if a GO term is still enriched after removing the genes belonging to a more significant parent term.

Performing actual conditional re-testing can be computationally expensive if done for all pathways one by one (would require new MVN simulation minus one gene's effect). But if we just want to qualitatively identify such cases, we can rely on logic: - If a single gene is extremely significant (p ~ 1e-8) and is present in multiple pathways, any pathway with that gene will get a big boost (since one gene with 1e-8 can make Fisher/Stouffer etc highly significant). - We can measure "pathway dependence on a gene" by something like: compute pathway p with and without that gene. To approximate without heavy sim: One can approximate by excluding that gene's p from the combination. For Fisher, for example, remove its $-2\ln(p)$ contribution (and reduce df by 2). For Stouffer, remove its Z from sum. For ACAT, recompute sum without that term. Then recompute p-value (could approximate or quickly simulate the smaller set). - This is doable for top gene contributions. At least for identification: if a pathway's p goes from 1e-8 to 0.2 after removing one gene, clearly that pathway was essentially that gene. If it stays ~1e-4 even after removing top gene, then multiple genes contribute. - Sun et al.'s step-down method basically did that systematically and found many pathways were solely reliant on a few genes [63] [42].

1. **Prune pathways with extremely high overlap** (preemptive approach):
2. One could decide to remove one of a pair of pathways that overlap >X% to reduce testing burden (some analyses do that to have more independent sets). But this is tricky: you might remove one that actually had something unique in its extra genes.
3. Alternatively merge them into a "meta-pathway", but then interpretation gets fuzzy.
4. Usually better not to prune before, but to interpret after clustering or step-down.
5. However, extremely broad "trash-can" pathways (like GO: "cellular process" with 1000 genes) often overlap with more specific ones. If a broad category is significant, often many subcategories will be as well. Conversely, sometimes broad might not while specific is.

6. We can note if a very broad pathway (like Reactome "Metabolism" with 800 genes) is significant, it's hard to act on that. One could check subpathways for narrower insight. Possibly not needed if focusing on significant sets (which often are moderately sized).

7. **Flagging single-locus or gene family effects:**

8. Sometimes a pathway is essentially a list of very similar genes (like a gene family). If one gene family member has a strong GWAS hit (like HLA-B in MHC class I pathway), then the entire pathway of similar genes (like "MHC class I protein complex" containing HLA-A, B, C, etc.) might show multiple moderately significant genes (since often correlated phenotypes or LD in region). But if they are all in one locus, that's really one locus effect affecting all those genes (maybe via LD or pleiotropy).
9. Another scenario: "Pathway" = gene cluster on a chromosome. That's not common in curated sets, but some GO terms are cluster-based (like HOX cluster genes).
10. If a pathway's member genes are physically clustered and all show association, it could be one genetic signal (like a big LD region).
11. To address, one can do a conditional analysis on that locus or note that all genes in that pathway come from region X and likely represent one association signal.

12. E.g., a pathway might have 5 cytokine genes all in a cluster and one SNP in that cluster affects all five, giving each gene moderately significant p. The pathway looks enriched (5 moderately low p's), but it's one variant affecting multiple genes (maybe via shared regulation or LD).
13. Distinguishing that from true multi-gene effect might require fine-mapping or conditional GWAS, which is deep analysis. But at least acknowledging region-based inflation: "We observed multiple chemokine genes on chr17 in one pathway with moderate p (~1e-3 each). They are closely linked, suggesting a single locus drives that enrichment; pathway significance disappeared after conditioning on the top SNP in that locus."

Implementing in CATFISH pipeline:
- We can incorporate a post-processing step: 1. **Identify "index" genes that drive multiple pathways.** For each significant pathway, list its top 1-3 genes by gene p. Count how often each gene appears among those top genes in all significant pathways. Genes like FGFR2 would appear in dozens of pathway lists if it's very significant [38]. Mark those pathways as gene-driven. A more quantitative measure: for each pathway, check the gene with smallest p, see what fraction of the pathway's weight of evidence it accounts for. If removing it drops –log10(p) by, say, > 50%, that pathway is mostly single-gene. 2. **Merge pathway results by gene/ locus clusters.** If 10 significant pathways are all due largely to the same set of 2-3 genes, report that as one finding ("Pathways related to FGFR2 signaling were enriched, driven by FGFR2 itself" [38]). 3. **Single-gene or small pathways:** Exclude or at least denote them. For example, "Pathway X (which has 2 genes) is significant because one gene is highly associated (p=…); this is effectively a single-gene hit, not a broad pathway effect." 4. **Highly overlapping pathways cluster:** Use Jaccard similarity among significant pathways to cluster. We can pick a representative (maybe the one with lowest p or the most general one). In results table, we could annotate clusters (like cluster A, B, etc). 5. **Pathway overlap reduction methods in literature:** There is a concept of controlling for gene overlap by permutation – e.g., one approach is to build a null where gene-level hits are randomly distributed across pathways given pathway sizes, but that's complicated. Another is using conditional hypergeometric tests (like test pathway B only on genes not in pathway A if A is known to be significant and overlapping). That's analogous to forward selection: pick the most significant pathway, then remove its genes from consideration and test others on residual data. This is essentially what step-down does in parametric form [62] [63]. - If time/performance allowed, one could do: find top pathway, then set the genes in that pathway to null (p=1) in the data, re-run CATFISH for remaining pathways. See if new ones pop or which drop. This is heavy but doable with the simulation approach if iterative. - But often simply noting it qualitatively suffices in discussion.

Example from Sun et al.: They reported after step-down that pathways dependent on one gene (FGFR2) were sifted out, leaving those with multi-gene support [63] [42]. They argued the latter are more biologically meaningful.

In our analysis: - We should definitely highlight if a single gene appears to cause a large portion of our pathway hits (like a well-known GWAS gene present in many sets). - Also mention if multiple significant pathways share a high proportion of genes (like immune pathways often share many immune system genes – so if one is up, many similar ones are up). - One tactic: run an *over-representation analysis on the set of significant pathways to see which genes are most frequent.* That ironically is like doing the reverse: find which genes occur in a lot of the significant sets. Those likely are the culprits driving many signals.

Finally, consider pathway pruning to present results: - Some publications present only the most specific pathways among a set of nested ones. E.g., if "Apoptosis" and "Intrinsic apoptotic signaling via p53" are both significant, they might report only the latter if it's significant and more specific. - One can choose to present either higher-level categories (if multiple specific ones are redundant) or specific ones (if broad ones are too generic). - Possibly use hierarchical ontology (for GO) to reduce redundancy by picking a representative node.

Single-gene proxy pathways detection: - This can be automated: if pathway size is <= 3 and it's significant, check the gene p's. Likely one gene is super significant; you can label it. - Could also detect if a pathway's name itself is a gene or family (like "TP53 network" – if that pathway's basically built around TP53 gene). - If a pathway has name like "GeneA Pathway" and indeed geneA drives it – trivial. E.g., "KEGG: Insulin signaling" might heavily rely on INS gene if that gene had a variant.

Overlapping gene sets and conditional analysis: - If two pathways share, say, 90% genes, then essentially they will have identical test statistics or highly correlated. For interpretation, treat them as same signal. For FDR, they count as separate tests though, which is one reason multiple testing burden is higher. Some approaches (like a *hierarchical test procedure*) could adjust for that, but it's complicated. Instead, we rely on detection then consolidation in interpretation.

Conclusion for Section 6:
It's crucial for the credibility of pathway findings to show that we acknowledge these redundancies. We should: - **Flag pathways that are driven by single genes or the same gene cluster.** For example, if **multiple top pathways contain BRCA1 and BRCA2 genes, and those are significant,** then all those pathways might simply reflect "DNA repair via BRCA1/2" association. We should condense that insight to "DNA repair pathways are enriched due to BRCA1/2 risk variants." - **If possible, perform a step-down analysis**: remove top genes and retest to see which pathways remain. If we can't fully retest, at least qualitatively describe it.

Sun et al.'s strategy is a gold standard approach [62] [63]. They argue it yields pathways that are more replicable (because if one gene was propping it, another dataset might not have that gene, then it fails to replicate). Multi-gene pathways have better chance if multiple genes truly moderate associated.

For CATFISH's output, we should plan to present results in a non-redundant way: - Possibly provide a table of significant pathways but grouped by shared genes or flagged with notes ("contains gene X which drives significance, see also pathway Y,Z with same gene"). This transparency helps readers not overinterpret the sheer count of pathways.

To manage pathway overlap systematically: - We could cluster by gene overlap and then do something like select the pathway with lowest p in each cluster as representative. Or we could list them but indicate they form a cluster. This is more a presentation decision.

Pathway pruning example: FGFR2 example: 86 of top 100 pathways contained FGFR2 [38]. Instead of listing all 86, one should just say "FGFR2 appears in many top pathways (Ear morphogenesis, etc.), indicating that the strong association of FGFR2 (p<1e-300) [64] is inflating numerous pathways that include it. When FGFR2 is excluded, those pathways are no longer significant [39]." And then focus on any pathways that remain significant beyond FGFR2. They literally did that: ear morphogenesis with FGFR2 was top100, without FGFR2 minimal association [39].

So in our context, after obtaining results we should: - Identify any such gene or small set dominating results, - Possibly re-run pathway p without it for confirmation, - Then report final pathways focusing on truly enriched ones.

Conditional test beyond single gene: - If a pair of genes always co-occur (like gene family always appears together), could try remove both and test. But diminishing returns – likely if it's more than 2, it's just a chunk of pathway, which the test would drop a lot.

Pathway cluster correlation and FWER: - Note: extremely overlapping sets mean the number of independent tests is less. We discussed using effective test count earlier. If correlation is extreme, Bonferroni is overly conservative. But usually we handle that through FDR or through acknowledging clusters of results (which is more interpretation than formal correction).

Wrap up Section 6:
We will implement at least two aspects in writing: (a) mention of single-gene and locus-driven pathways with how to address them, and (b) mention of clustering/conditional analysis to parse multiple hits.

7. Simulation and Benchmarking Designs for CATFISH Validation

To have confidence in CATFISH's statistical performance, we need to **validate its type-I error control and assess its power** in realistic scenarios through simulation and benchmarking. This involves creating synthetic datasets with known pathway architectures and also testing CATFISH on reference datasets or via cross-validation.

Type-I error (false positive control):
CATFISH's omnibus p-values are calibrated by simulation under the null, but we should verify that in practice: - Under null, ~5% of pathways have $p<0.05$, ~1% have $p<0.01$, etc., even at the tail end like $p<10^{-4}$ should be around 0.01% of pathways if none are truly associated. This can be evaluated by: - **Null simulations genome-wide:** Generate phenotype data with no genotype-phenotype association. For example, permute case/control labels or simulate a quantitative trait by randomly shuffling or sampling from a normal with no SNP effect. Use the real genotype matrix (or summary stats with null effect but real LD structure). - Then run the entire CATFISH pipeline (compute gene p's via MAGMA, then gene-set p's, then combine). - Count how many pathways are flagged at various significance levels. Ideally, these should roughly match the nominal α levels. Particularly test the extreme threshold that we intend to use (e.g., if threshold is $5e-6$ for significance, in, say, 100 replicates of null, how often do we get any pathway with p <5e-6? It should be ~5% of the replicates if threshold is correct FWER). - If any inflation is observed (more false positives than expected), adjust calibration or identify cause (likely gene correlation matrix issues). - Pascal did something similar with random phenotypes and produced QQ-plots showing uniform distribution of pathway p's after their adjustments [28] [29]. We should likewise be able to demonstrate a flat QQ-plot for pathway p under null, even at tail. - If doing 1000 null simulations of the whole genome is too heavy, we can approximate by simulating gene-level stats from MVN as we already do. Actually, we can use the same MVN apparatus to simulate many "studies": sample gene-level Z's for all genes from MVN(0,R_global) (if we have an approx global gene correlation across genome, which is block diagonal by chromosome presumably). Alternatively, simulate chromosome by chromosome and merge results (since different chromosomes are independent under null). - Then run the pathway combination on each simulation's gene p's and see distribution of min p or count of low p's. - This is heavy but feasible with enough computing if parallelized. A smaller number of replicates (like 100 or 200) can at least show if anything egregious is wrong (like if distribution is way off). - Also ensure that correlation structure is

correctly handled. If issues appear (like slight inflation due to gene overlaps), consider using gene-fusion or refining calibration.

- **Permutation vs MVN simulation:** Perhaps confirm that doing permutation of phenotypes (thus preserving LD and any structure) yields similar results to our MVN approach for null. If they match, it's strong evidence our MVN modeling is accurate.

- **Long-tail estimation:** Type I at very small α (like 1e-6) is harder to directly validate because we would need many replicates. But one can do something like extrapolate or use extreme value theory. Possibly run fewer replicates and fit a distribution to min p etc.

- Specifically, test that *FWER is controlled at the desired level.* For instance, to claim in publication that we have genome-wide significance at $p<5e-6$ controlling FWER 5%, we should show in null simulations that the chance of any pathway having $p<5e-6$ is ≈ 0.05. If it came out as 0.08 (meaning threshold too liberal), we might adjust threshold or at least calibrate with a slight tweak. Our earlier discussion suggests maybe effective number of tests could allow a slightly higher threshold than Bonferroni; but to be safe likely stick to a bit conservative side.

- Another aspect: *Family-wise false discovery proportion.* Even at FDR level, we can check that in null data BH yields ~0 discoveries (some fluctuations). But more importantly, in alternative scenarios, does BH approximate the actual false discovery proportion.

Power under diverse architectures:
We want to ensure CATFISH indeed improves power (detects signals) in scenarios from single-gene to polygenic pathways. We should simulate various alternatives: - **Architecture parameters:** Consider: - Number of causal pathways (maybe 1 or a few in each sim). - For each causal pathway: how many causal genes? how large effects? - Spread of effects: concentrated in one gene vs spread across many vs intermediate. - Overlap between pathways: possibly have overlapping pathways, to see if CATFISH can handle picking them out or if it flags both (which it likely will, then we see how step-down might differentiate). - Also difference in gene-level architecture within pathway: e.g., one gene could have multiple independent SNP effects vs each causal gene only one SNP cause, etc. But since we feed gene p's, we could abstract that away by directly setting gene p's as input maybe (though better to simulate from SNP level to incorporate correlation properly).

- **Example simulation scenarios:**
- **Single gene pathway effect:** Only one gene in the pathway is truly associated (carries all effect). This tests if CATFISH picks it up (should be through minP or ACAT).
  - Vary that gene's effect size so that the gene-level p is around some threshold (to see where detection kicks in).
  - We expect minP and ACAT heavy tests to excel, others not as much. The omnibus should do well because minP-O will catch it.
  - Check type I: if the effect is moderate (p borderline), do we inadvertently inflate others? Probably fine.
- **Few gene (sparse) effect:** e.g., 3 out of 50 genes in pathway have modest effects (each yields gene p ~0.001). No single one is genome-wide significant, but collectively they should trigger Fisher/ Stouffer.
  - Does CATFISH detect the pathway? Likely yes via Fisher or ACAT.

- Which of the five tests triggers: possibly soft TFisher or Fisher will produce lowest p.
- Compare to if using simpler approach like hypergeometric (select genes p<0.01 vs expectation). Our method likely more optimal because it uses actual p values.
- If possible, compare to MAGMA's own competitive gene-set test if available (though MAGMA's competitive test is different null).

• **Dense effect:** e.g., 50% of genes have a small effect (each gene p ~0.05 or 0.01). Each alone wouldn't stand out but together the whole set is clearly shifted.
  - Fisher/Stouffer should nail this; ACAT might also combine enough weight to show significant. BH might also find a portion of those genes and maybe the pathway by hypergeometric if threshold set not too strict.
  - We check CATFISH finds it at pathway-level while perhaps no gene is individually significant after correction.

• **Mixed directions**: Genes have effects in different directions (for case-control, some OR >1, some OR <1). Non-directional tests should still capture overall association (because they use two-sided p). Directional test would fail here.
  - Confirm that ignoring sign (which we do) still yields detection. Likely yes because each gene's two-sided p is low. If half genes protective, half risk, the pathway is still enriched in low p's.
  - Just ensure that not doing a directional test doesn't miss something – it shouldn't, it just doesn't test coherence of sign.
  - For interest, one could calculate directional Stouffer and see it ~0 (if cancellation) to confirm that a directional method would have missed it (which is an argument *against* using directional as default).

• **Multiple overlapping pathways:** e.g., define two pathways that share some genes. Simulate causal genes such that:
  - Both pathways have signals, but they share a key gene that is causal. We might see both pathways significant. See if our method flags both. It will. Then test if step-down removal of that shared gene would drop both maybe (depending on if they had other signals).
  - Or one pathway is a subset of another: e.g., Pathway A has 5 causal genes (out of 50). Pathway B is those 5 genes only (subset). In reality, B is "within A". Both should be significant. Ideally, we'd identify that B is basically highlighting those same genes as A in a more focused way. FWER viewpoint: B being smaller might have more dramatic p (fewer genes diluting).
  - See if any advantage or issue: e.g., maybe minP-O combining the same tests yields both extremely significant (since they share signals), which is fine – just a scenario to interpret.

• **Null pathways vs causal pathways**:

  - We want to measure power: e.g., simulate 1000 sets with say 10 having some effect, see how many of those 10 CATFISH detects (power), and how many of the other 990 it falsely flags (type I).
  - Vary effect sizes or # of causal genes to produce power curves. For instance: fraction of causal genes needed to reach certain detection rate at given sample size.
  - These could be computationally heavy but at least do a few representative points.

• **Performance relative to single-SNP GWAS:** Show that pathway analysis can find signals that SNP-level might miss. For example:

• If each gene in a pathway has at least one SNP with p ~1e-4, none is GW-significant alone, but 10 such genes in one pathway yields pathway p ~ something like 1e-6 or smaller (depending combination method). Then the pathway is discoverable even though no single SNP or gene was at 5e-8. This demonstrates the *value-add of pathway analysis* – which likely will be one highlight in a publication.

- We could simulate such scenario and indeed show CATFISH finds the pathway while a standard approach (just single SNP or gene tests with correction) finds nothing.

- Real example: They had in Pascal paper demonstration on Crohn's disease where gene-level analysis found more than SNP-level [65]. We can similarly demonstrate at pathway level using some known polygenic trait that certain pathways are identifiable only via combined evidence.

- **Robustness tests:**

- *Misspecification of correlation:* If our R_S is off, does calibration hold? (We partly test that with null).

- *Missing heritability scenario:* If truly everything is polygenic at small effect all over, we might see no specific pathway pass significance, but maybe many pathways with modest enrichment (like in height, the entire genome has small effects). FDR might highlight broad categories (like "whole genome" type categories) if polygenicity is widespread. Check that we don't get false positives in a scenario where everything has tiny effects (which is basically null in discrete discovery terms).

    - Possibly simulate a polygenic background: all genes have a teeny effect such that no pathway is truly "enriched" more than any random group. Check that CATFISH doesn't produce spurious hits beyond expected false positives. That is tricky because with heavy polygenicity you might see slight deviations (random high clusters).
    - Essentially similar to null but with slight correlation structure in outcomes beyond LD.

- **Comparative power:** If possible, compare CATFISH to other pathway methods:

- e.g., apply a simple competitive enrichment test like MAGMA's gene-set test or hypergeometric test on the same simulated data, see if CATFISH's self-contained approach picks up signals with equal or better sensitivity and at controlled false positive rate.
- For known gene-set methods (like GSEA implemented via permutation, or logistic regression of phenotype on gene scores), maybe not easily available to test on summary-level data, but one might implement a competitive test by permuting case labels in simulation to evaluate significance of gene sets (that's essentially how many older methods do).
- If show that CATFISH yields similar or better power than say hypergeometric, plus doesn't need permutations for each test, that's a selling point.

Edge cases for validity: - Pathways at extremes of size: Very large or very small sets. - Ensure type I is okay for large sets (with many genes, there's more chance to get a small p by chance via Fisher or minP if so many draws – but calibration via MVN of correlated genes should handle it). Pascal specifically handled large sets by adjusting for gene correlation and size [26] [28]. Check that in null simulation large sets don't have systematic inflation. They likely not if gene correlation considered. - For tiny sets, we typically would exclude <5 genes because results can be noisy or trivial. If included, calibrations for sets of size 1 are exact, size 2-3 can calibrate by sim easily. Null should be uniform because we do simulation for each set individually.

Implement advice and improvements from sim: - If we find, e.g., that using ACAT-O as omnibus yields slightly conservative or liberal combined p relative to some known approach, we might adjust (but likely it's fine). - If simulation shows that calibrating only omnibus and not each test leads to slight misestimation of component tests, it may not matter for final, but could mention adjusting each individually if needed. - If any anomaly found (like pathway with many overlapping genes resulting in difficulty), might incorporate gene-fusion or mention as caution.

Preparing for publication: We should have evidence: - Type I error controlled (maybe with a QQ-plot or table from null sims) [28] . - Simulation showing power in certain scenarios (maybe a figure with power vs number of causal genes, or examples). - Possibly one real-data demonstration: e.g., apply CATFISH to a known dataset with a known pathway, see if it finds it. If none known, rely on sim.

Corner: Pathway Redundancy & type I: If same gene appears in many sets, in a single simulation replicate that gene might generate multiple false-positive pathways if it has an extreme null fluctuation. E.g., under null, gene A might by chance have p=1e-4 (like from random effect), that could make a bunch of pathways containing A have p ~ maybe a bit small too. That could inflate family-wise error because one null fluctuation yields multiple hits. But our FWER is typically counting any hits, and if one gene gave multiple, it still counts as one event of FWER (because at least one path got flagged, and indeed many did). - It means tests are not independent; Bonferroni accounted as if independent which is overly conservative, but simulation calibrations should account for that somewhat. - So e.g., in null simulations if one gene happens to have an unusually low p, it might make cluster of pathways all crossing threshold in that replicate. That means the "any pathway < threshold" event can be correlated. But that's fine, it just means if threshold too high, many will cross at once boosting FWER beyond 0.05. Our simulation should catch that and we set threshold accordingly. - It again highlights the need to cluster in interpretation because a single null gene blip could produce many false pathways. But as long as we count it as one false discovery event for FWER (which we do, any is any), it's okay for controlling FWER. But for FDR, if that happened, BH might call many significant (so FDR procedure's assumption of independence is violated, potentially making actual FDR > nominal, though BH is robust under positive correlation typically, and here correlation is positive because a gene being low ensures path low). - We might test via simulation that the BH procedure still roughly controls FDR under null or modest signals. BH controlling FDR is proven under independence or certain dependency, but extreme overlaps might slightly break conditions (like that scenario all those pathways share that gene - they are highly positively dependent, which actually is one case BH still is okay or slightly conservative, BH is worst under negative dependence which doesn't occur here).

Simulated threshold calibrations vs analytic: - We likely rely on simulation to calibrate each pathway's p, which inherently accounts for gene overlaps within that pathway (self-contained). - But across pathways, we rely on multiple testing procedures (FWER or FDR). - We might simulate to confirm that using e.g. Bonferroni at 5e-6 indeed yields ~5% global false positive under global null. - If we wanted, we could incorporate an *empirical null max distribution* to refine that threshold as earlier. Might or might not, but at least we would know how conservative Bonferroni is given overlaps in the tested dataset.

All these designs ensure method validity. We can then confidently prepare the method for publication, stating: - We performed extensive simulations verifying type I error control at stringent levels (no inflation up to e.g. p=1e-6) [28] . - We demonstrated power in scenarios with varying proportion of causal genes, showing CATFISH's omnibus (especially ACAT-O) is robust across sparse and dense alternatives [1] [11] . - In comparison to simpler methods (like hypergeometric or single-gene tests), CATFISH finds pathway signals that would be missed at stringent thresholds, thus offering a gain in discovery. - Also, mention if any particular best practices gleaned, e.g., "Use at least B=1e6 null simulations for each pathway to accurately estimate p down to 1e-6" – basically guidelines for how to implement to achieve accuracy. - Emphasize controlling for multiple testing across pathways (e.g., recommending FDR 5% for discovery, and using empirical null for FWER if wanting high confidence).

Finally, implementing these suggestions (like merging overlapping gene sets, etc.) will make results more digestible and reduce false or redundant claims.

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1 2 3 19 Aggregated Cauchy Association Test (ACAT)
https://lawlessgenomics.com/topic/acat

4 5 6 16 17 18 41 (PDF) The harmonic mean p -value for combining dependent tests
https://www.researchgate.net/publication/345720605_The_harmonic_mean_p_-value_for_combining_dependent_tests

7 8 9 32 33 34 Combining dependent P-values with an empirical adaptation of Brown's method - PubMed
https://pubmed.ncbi.nlm.nih.gov/27587659/

10 11 12 13 14 20 21 22 23 24 38 39 40 42 62 63 64 Powerful gene set analysis in GWAS with the Generalized Berk-Jones statistic | PLOS Genetics
https://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.1007530

15 TFisher: A powerful truncation and weighting procedure for ...
https://projecteuclid.org/journals/annals-of-applied-statistics/volume-14/issue-1/TFisher--A-powerful-truncation-and-weighting-procedure-for-combining/10.1214/19-AOAS1302.full

25 31 35 36 65 MAGMA: Generalized Gene-Set Analysis of GWAS Data | PLOS Computational Biology
https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1004219

26 27 28 29 30 37 43 44 47 48 49 50 51 52 53 54 55 56 57 Fast and Rigorous Computation of Gene and Pathway Scores from SNP-Based Summary Statistics | PLOS Computational Biology
https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1004714

45 46 A simplified protocol for performing MAGMA:H-MAGMA gene set analysis utilizing high-performance computing environments.pdf
file://file-UTii4V3G4hakky35dcoMkB

58 59 60 61 Fast set-based association analysis using summary data from GWAS identifies novel gene loci for human complex traits | Scientific Reports
https://www.nature.com/articles/srep32894?error=cookies_not_supported&code=82cff35b-ec3c-401c-a34a-7c492bced6b3